

XAI-SDN: An explainable entropy-guided machine learning framework for real-time DDoS detection in software defined networks

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ABSTRACT Software-defined networking concentrates forwarding decisions in a logically centralized controller. That simplifies management and creates a single point of failure: a distributed denial-of-service campaign that saturates the control channel can disable forwarding domain-wide. Machine learning detects such traffic well. An operator who cannot see why a flow was flagged will not authorize autonomous mitigation. We present XAI-SDN, a controller-embedded detector. It augments exported flow statistics with eight windowed Shannon entropy features at expected amortized constant cost per update, classifies with a Random Forest, and attributes alerts with exact Shapley values under an explicit explanation budget. Under a chronological split of the CIC-DDoS2019 SYN partition, where every training flow precedes every test flow, it attains 99.76% accuracy and 97.88% macro F1 over 1,077,238 held-out flows at a 0.095% false positive rate. A stratified random split of the same data yields 99.95% macro F1. We report both, because the protocol moves the headline metric further than any architectural choice examined here. Attribution, not classification, is the binding cost: explaining every positive prediction cuts sustained throughput from 35,715 to 661 flows per second, whereas one exact attribution per attack episode restores 35,652, with every alert still resolvable to an attribution. A live OpenFlow 1.3 control plane under Mininet supplies the deployment figures: controller CPU and memory, detection latency, the offered rate at which one controller saturates, and the flow-rule aggregation bounding flow-table occupancy during a flood. Transfer to an unseen corpus fails without target-domain labels, a limitation reported rather than bounded away.

INDEX TERMS DDoS detection, evaluation methodology, explainable AI, network security, random forest, Shannon entropy, SHAP, software defined networks

I. INTRODUCTION

SOFTWARE-defined networking (SDN) separates the data plane from the control plane and concentrates forwarding decisions in a logically centralized controller [1], [2]. Enterprise and cloud operators adopt the architecture because a single control point simplifies reconfiguration, admits network-wide visibility, and makes traffic engineering programmable. The same concentration creates an obvious target. A distributed denial-of-service (DDoS) campaign directed at the controller, or one that simply generates enough table-miss events to saturate the control channel, can disable forwarding across the entire domain.

DDoS traffic takes several forms: UDP amplification, TCP SYN flooding, ICMP sweeps, and application-layer attacks such as HTTP floods and Slowloris [3], [4], [5]. Machine learning detects these patterns well, but most deployed detec-

tors are opaque. An operator who cannot see why a flow was flagged cannot adjudicate a borderline alert, cannot defend an automated mitigation to an incident review, and in practice will not authorize the system to act autonomously [6], [7], [8], [9].

Three requirements therefore apply at once: accurate detection, processing cheap enough to keep pace with the control channel, and decisions an operator can inspect. Systems in the literature typically satisfy two [10], [11]. The difficulty is not that any one requirement is hard in isolation but that the third is usually paid for by sacrificing one of the others, and the size of that payment is rarely measured.

This paper measures it. We present XAI-SDN, a controller-embedded detector that augments CICFlowMeter flow statistics with windowed Shannon entropy features maintained at expected amortized constant cost per update, classifies with

a Random Forest, and attributes alerts with exact TreeSHAP under an explanation budget: every admitted alert receives an attribution computed for that specific flow, and every alert the budget declines carries a reference to the exact attribution of its episode representative. Section III-B fixes the distinction and Section VII-K measures what it buys. The contribution is less the combination itself than the operational accounting that surrounds it, and the methodological care that the accounting turned out to require.

Our contributions are as follows.

- **A numerically characterized streaming entropy engine.** We give the update recurrence the implementation actually executes, state its complexity as expected amortized $\mathcal{O}(1)$ per admit/evict pair under explicit data-structure assumptions rather than as an unqualified bound, and measure the floating-point drift it accumulates over 10^7 updates together with the number of classifier decisions that drift changes.
- **A formally defined and budgeted explanation trigger.** Attributing every positive prediction is untenable when almost all traffic is hostile. We define the trigger once, measure the throughput collapse it causes, and quantify the throughput recovered by episode-level deduplication, a token-bucket budget, and off-path asynchronous attribution, reporting the resulting throughput versus explanation-coverage frontier.
- **A control-plane cost accounting.** Using an OpenFlow 1.3 testbed in which the detector runs inside a live controller process, we report controller CPU and memory, end-to-end detection latency, the control-channel rate at which a single controller saturates, and the flow-rule aggregation and eviction policies required to keep a bounded TCAM from overflowing during a flood.
- **An evaluation-protocol audit.** We show that split methodology, not model capacity, accounts for most of the difference between the headline figures usually reported on this benchmark and the figures a chronological protocol yields, and we report both so the gap is visible. We pair this with a leakage audit, a collinearity and attribution-stability analysis, cross-vector and cross-day evaluation, and cross-dataset transfer to InSDN and CIC-IDS2017 with confidence intervals on the error rates.

We make no claim that XAI-SDN dominates prior work on every axis. The claim is narrower: an entropy-augmented tree ensemble with exact attributions occupies an operating point where detection quality, control-plane cost, and per-decision transparency can all be stated in measured numbers, and this paper states them, including where they are worse than the literature would lead one to expect. Two of those numbers concern the augmentation this system is named for, and they point in opposite directions. Under the chronological protocol the eighty exported flow statistics alone match the full 88-dimensional vector, so the entropy features do not improve in-distribution detection on this corpus. Under every shift for which we have data they do, provided all eight can

be computed. Section VII-F reports both, together with the control that locates the cause of the first and the corpus that bounds the second.

Table 1 lists the symbols, abbreviations, and acronyms used throughout.

The remainder of the paper is organized as follows. Section II reviews related work. Section III defines the threat model and formalizes the detection problem. Section IV describes the XAI-SDN architecture and Section V its mathematical foundation. Section VI details the experimental setup. Section VII reports detection, latency, and explainability results. Section VIII presents the ablation and baseline comparison. Section IX discusses deployment considerations and limitations, and Section X concludes.

II. RELATED WORK

Work relevant to this paper falls into four groups. Entropy-based detectors established that distributional summaries expose flooding that per-flow statistics miss, but they act on fixed thresholds rather than learned boundaries. Learned detectors improved accuracy at the cost of transparency and, as several methodological critiques argue, of evaluation rigor. Explainable AI supplies the missing transparency, rarely with an accounting of what an explanation costs on the alert path. A fourth group, developed outside networking, builds nonlinear dynamic indices for abnormal-state detection and offers a vantage point on why distributional features work at all.

A. ENTROPY-BASED DDOS DETECTION IN SDN

Shannon entropy summarizes how concentrated a traffic attribute is over a window, and concentration is what a flood produces. Nychis et al. [12] characterized the correlation structure of entropy time series across several traffic distributions, and Lakhina et al. [13] showed that entropy exposes anomalies volume-based metrics do not register, establishing it as a standard anomaly-detection primitive [5]. In SDN, Mousavi and St-Hilaire [10] showed that a threshold on destination-address entropy, evaluated at the controller, detects flooding early enough to limit its impact, and Lall et al. [14] developed streaming estimators that make entropy tractable at line rate over sliding windows.

The relationship these methods exploit is statistical rather than deterministic. Low source-address entropy may indicate a concentrated flood, but the mapping depends on which attribute is summarized, how long the window is, what the background traffic happens to be, and how the attacker distributes its sources. An adversary who rotates spoofed addresses to hold the per-window distribution near uniform weakens the signal deliberately, a point Section IX returns to. Fixed thresholds also cannot represent interactions between distributional and flow-level evidence. XAI-SDN keeps the streaming entropy primitive but treats its output as features for a classifier rather than as a decision rule, at expected amortized $\mathcal{O}(1)$ cost per update under the assumptions stated in Section V.

TABLE 1. Notation, symbols, and acronyms.

Symbol	Meaning	Symbol	Meaning
i, j	flow index, feature index	W	sliding window of flow keys
f_i	i -th flow at the controller	N	window capacity, in flows
$\mathbf{x}_i$	feature vector, $d = 88$	n_k	multiplicity of key k in W
$y_i, \hat{y}_i$	true, predicted label	$\mathcal{K}(W)$	distinct keys present in W
$g_\theta(\cdot)$	scoring function	$H(\cdot)$	Shannon entropy, in bits
s_i	DDoS score of f_i	S	running $\sum_k n_k \log_2 n_k$
τ	detection threshold	$\psi(m)$	$m \log_2 m$, with $\psi(0) = 0$
$B(f_i)$	budget admission decision	T	trees in the ensemble
$\mathcal{F}$	feature index set $\{1, \dots, d\}$	L, D	max leaves, max depth
ϕ_{ij}	contribution of feature j to s_i , Eq. (6)	$\mathcal{C}$	{Benign, DDoS}
ϕ_i	exact TreeSHAP attribution computed for flow f_i	$\sigma(f_i)$	attack-episode signature
ϕ_0	TreeSHAP base value	$\rho(i)$	episode representative of f_i
$\mathbf{a}_i$	attribution the alert record for f_i carries, Eq. (4)	Acc	accuracy, Eq. (7)
TP, FP	true, false positive count	P_c, R_c	per-class precision, recall
		F1 _c	per-class F1
		TN, FN	true, false negative count
Acronym	Expansion	Acronym	Expansion
AP	average precision	ROC	receiver operating characteristic
AUC	area under the curve	SD-IoT	software-defined Internet of Things
DDoS	distributed denial of service	SDN	software-defined networking
FNR, FPR	false negative, false positive rate	SHAP	SHapley Additive exPlanations
IAT	inter-arrival time	SOC	security operations center
OXM	OpenFlow extensible match	TCAM	ternary content-addressable memory
PR	precision-recall	VIF	variance inflation factor
RF	Random Forest	XAI	explainable artificial intelligence

B. MACHINE LEARNING FOR SDN DDOS DETECTION

Braga et al. [15] showed early that lightweight learning over OpenFlow counters suffices to detect flooding at the controller. Deeper architectures followed, including deep neural classifiers [16] and stacked-autoencoder detectors [17], trading inference cost and transparency for accuracy. Recent work extends the progression into graph and sequence models: E-GraphSAGE [18] learns over the host-interaction graph, and FlowTransformer [19] applies encoder attention to flow sequences. Bhayo et al. [20] carry the approach into software-defined IoT, and Jain et al. [11] survey the field. Section VIII evaluates a representative of each family under one protocol on the same hardware, because published numbers are not comparable across papers.

Benchmarks have shaped the field as much as architectures have. CIC-IDS2017 [21] and CIC-DDoS2019 [4] supply labeled flow-level captures at a scale that supports supervised training, and InSDN [22] contributes traffic from an actual SDN testbed. Their convenience has a cost. Intrusion-detection surveys [23] and the critiques of Sommer and Paxson [24] and Arp et al. [25] document how easily results on such corpora are inflated: random splits scatter near-duplicate flows across both partitions, temporally segregated classes let a model key on capture phase instead of attack behavior, and synthetic generators leave per-feature signatures that separate the classes without any attack semantics being learned. Section VII reports what each of these contributes on the partition we evaluate.

C. EXPLAINABLE AI FOR INTRUSION DETECTION

Guidotti et al. [7] organize the space of black-box explanation methods, and Doshi-Velez and Kim [26] set out what rigorous evaluation of interpretability would require [6]. Two attribution methods dominate practice: LIME [27], which fits a local surrogate, and SHAP [28], which yields additive attributions with consistency guarantees. TreeSHAP [29] makes the latter exact and polynomial for tree ensembles, which is what makes per-alert attribution viable in a real-time path. Gaspar et al. [30] apply both to a multilayer-perceptron intrusion detector, and Al and Sağiroğlu [9] survey the current state of the area. We use an exact method rather than a model-agnostic approximation because Warnecke et al. [8] find exact attributions both more faithful and cheaper in this setting.

Attribution answers which features drove a decision. It does not say how much confidence the decision deserves, and the two are often conflated. Swaminathan et al. [31] develop Bayesian machinery for quantifying predictive uncertainty, which is the complementary capability an operator needs when a score falls near the threshold. Our detector emits a vote fraction calibrated only in the loose sense of being monotone in evidence, and Section IX identifies Bayesian treatment of that score as the natural next step.

D. NONLINEAR DYNAMIC INDICES FOR ABNORMAL-STATE DETECTION

A parallel line of work, developed largely outside networking, argues that abnormality is easier to see in the structure of a dis-

TABLE 2. Qualitative positioning. ✓ means the property is present and reported, × that it is absent or not reported. The columns are the axes on which this work differs: whether an attribution is produced for each decision, whether the cost of producing it is budgeted rather than incurred per positive prediction, whether control-plane cost is measured rather than assumed, and whether the evaluation protocol is audited.

Approach	XAI	Budget	Ctrl. cost	Audit
Mousavi and St-Hilaire [10]	intrinsic	n/a	×	×
Lall et al. [14]	intrinsic	n/a	×	×
Braga et al. [15]	×	n/a	×	×
Lo et al. [18]	×	n/a	×	×
Manocchio et al. [19]	×	n/a	×	×
Gaspar et al. [30]	post-hoc	×	×	×
XAI-SDN (this work)	exact	✓	✓	✓

tribution than in any individual observation. Chen et al. [32] construct a tensor Poincaré plot index for pumped storage units by decomposing an operational signal across time and frequency scales and extracting geometric indices from the resulting plots, on the argument that scale-agnostic summary statistics discard exactly what distinguishes a healthy machine from a degrading one.

A methodological parallel to entropy-augmented flow detection is evident. Both replace a raw per-observation statistic with a distributional summary over a window, on the premise that the abnormal condition shows in the shape of the distribution rather than the magnitude of any single sample. Both must then answer whether the summary tracks the condition of interest or an incidental property of how the data were collected. Section VII answers that for our entropy features directly, by recomputing them over a shuffled arrival order and measuring how much discriminative power survives.

E. POSITIONING

Table 2 places XAI-SDN against the approaches above. What distinguishes this work is not the components, which are individually standard, but the combination of a streaming distributional feature set whose complexity and numerical behavior are both characterized, an explanation mechanism whose trigger is defined and whose cost is budgeted and measured, and an evaluation conducted at realistic class imbalance with the control-plane cost accounted for rather than assumed.

III. THREAT MODEL AND PROBLEM FORMULATION

We set out here what the defender observes, what the attacker controls, and how the detection task is formalized. We also fix four terms that are easy to conflate, because the distinction between classifying a flow and explaining that classification carries most of the computational cost reported in Section VII.

A. THREAT MODEL

An attacker controls a botnet and launches a volumetric DDoS attack against an SDN-controlled network, flooding it with traffic and additionally generating packets that match

no existing flow rule, so that table-miss events saturate the control channel and consume controller CPU. The attacker may spoof source addresses and vary packet timing but has no direct control over the controller or the switches. An adaptive attacker who optimizes against the detector is outside this model and is treated as a limitation in Section IX-D. The defender observes the flow statistics switches export and must decide in real time whether a flow is malicious, and produce a justification for that decision.

B. TERMINOLOGY

Four terms are used with fixed meanings throughout, including in the abstract, in Algorithm 1 and in every table reporting cost. A *flow* f_i is one bidirectional flow record exported to the controller; every flow is classified, so the number of classifications equals the number of flows. A *positive prediction* is a flow whose score reaches the detection threshold. An *alert* is a positive prediction dispatched to the security operations center: in the configuration evaluated here the two coincide numerically, but alerting is a policy decision layered on classification and the concepts are distinct. A *per-decision explanation* is an exact TreeSHAP attribution vector ϕ_i computed for one specific flow f_i , generated only for alerts the explanation budget admits, so per-decision explanations are in general a strict subset of alerts.

The fifth term is the one this distinction forces, and we state it separately because conflating it with the fourth would overstate what the system delivers. Alerts are grouped into *attack episodes* by a signature $\sigma(f_i)$, and the first alert of an episode that the budget admits becomes its *representative*, written $\rho(i)$ for the representative of f_i 's episode. An alert the budget declines carries the representative's attribution by reference rather than one of its own. Equation (4) writes this as $\mathbf{a}_i$, the attribution the alert record actually carries, and the two cases it distinguishes do not carry the same guarantee. Only the admitted case is an exact attribution of f_i 's own score. The inherited case is exact for $f_{\rho(i)}$ and is an episode-level summary for f_i : it identifies the evidence that placed this episode over the threshold, not the evidence that placed this particular flow there. Section VII-K quantifies the trade and Section IX-D records the limit. We therefore never describe the system as computing an exact per-flow attribution for every alert; we describe it as computing one for every *admitted* alert, and as leaving every remaining alert resolvable to its episode representative.

Costs are reported against the quantity that drives them. Classification cost is per flow. Explanation cost is per generated explanation, never per flow and never per DDoS flow, because the number of explanations is a policy variable rather than a property of the traffic.

C. PROBLEM FORMULATION

Let f_i be the i -th flow observed at the controller and $\mathbf{x}_i \in \mathbb{R}^d$, $d = 88$, its feature representation: 80 CICFlowMeter statistics and 8 windowed entropy features. The detector learns a

scoring function $g_\theta: \mathbb{R}^d \rightarrow [0, 1]$, writes $s_i = g_\theta(\mathbf{x}_i)$, and applies

$$\hat{y}_i = \mathbb{I}[s_i \geq \tau], \quad (1)$$

with τ selected on a validation partition, never on test data, and on the precision–recall plane rather than through the false negative rate, for the reason Section VII-H gives.

Explanation is a separate, conditional step. Writing $B(f_i) \in \{0, 1\}$ for the admission decision of the explanation budget, an explanation is generated exactly when

$$\text{explain}(f_i) = \mathbb{I}[\hat{y}_i = \text{DDoS}] \cdot \mathbb{I}[s_i \geq \tau] \cdot B(f_i). \quad (2)$$

Setting $B \equiv 1$ recovers the unbudgeted policy in which every positive prediction is attributed, and Section VII-K reports what that policy costs when almost all traffic is hostile. An explanation is an attribution vector $\phi_i \in \mathbb{R}^d$ satisfying

$$g_\theta(\mathbf{x}_i) = \phi_0 + \sum_{j=1}^d \phi_{ij}, \quad (3)$$

so the evidence decomposes exactly across features and can be audited after the fact.

Equation (3) is a property of an attribution computed for $\mathbf{x}_i$, and it is the reason a budgeted alert and an admitted one must be labeled differently. With $\rho(i)$ the representative of f_i 's episode as defined in Section III-B, the attribution the alert record carries is

$$\mathbf{a}_i = \begin{cases} \phi_i, & B(f_i) = 1, \\ \phi_{\rho(i)}, & B(f_i) = 0. \end{cases} \quad (4)$$

In the first case $\mathbf{a}_i$ satisfies (3) with respect to $\mathbf{x}_i$. In the second it satisfies (3) with respect to $\mathbf{x}_{\rho(i)}$, and it does so with respect to $\mathbf{x}_i$ only in the degenerate case $g_\theta(\mathbf{x}_i) = g_\theta(\mathbf{x}_{\rho(i)})$, which the budget does not enforce and which flows differing in duration, byte counts, ports or window composition will not in general satisfy. An inherited attribution is therefore an audit reference to an exact decomposition of the representative's score, not an exact decomposition of this flow's score, and no claim in this paper treats it as the latter.

The design objective is to maximize macro-averaged F1 under severe class imbalance, subject to a per-flow latency budget set by the control channel and an explanation budget set by the operator.

IV. XAI-SDN FRAMEWORK AND ARCHITECTURE

XAI-SDN comprises five stages: flow-statistics collection, feature extraction, Random Forest classification, explanation generation, and alert dispatch (Fig. 1).

An entropy engine (Fig. 2) maintains the distributional features of Section V from a circular buffer of recent flow keys and a hash map of key multiplicities, at expected amortized $\mathcal{O}(1)$ cost per update.

Positive predictions that the explanation budget admits are attributed with TreeSHAP [28], [29], and the resulting payload is forwarded to the security operations center for operator review. Algorithm 1 states the procedure, including

Algorithm 1 XAI-SDN detection with a budgeted explanation trigger

Require: Flow batch $\mathcal{B} = \{f_1, \dots, f_n\}$; RF model $\mathcal{M}$; Tree-Explainer $\mathcal{E}$; window W ; threshold τ ; budget policy $\mathcal{P}$

Ensure: Label set $\mathcal{Y}$; alert set $\mathcal{A}$

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1:  $\mathcal{Y} \leftarrow \emptyset, \mathcal{A} \leftarrow \emptyset, W \leftarrow \emptyset, \mathcal{P}.\text{Reset}()$ 
2: for each  $f_i \in \mathcal{B}$  do
3:    $\mathbf{x}_s \leftarrow \text{CICExtract}(f_i)$   $\triangleright$  80-dim statistics
4:    $\text{AdmitEvict}(W, f_i); \mathbf{x}_e \leftarrow \text{RollingEntropy}(W)$   $\triangleright$ 
   Eq. (11)
5:    $\mathbf{x} \leftarrow [\mathbf{x}_s; \mathbf{x}_e]$   $\triangleright$  88-dim vector, Eq. (12)
6:    $s_i \leftarrow \mathcal{M}.\text{Score}(\mathbf{x})$   $\triangleright$  Eq. (5)
7:    $\hat{y}_i \leftarrow \mathbb{I}[s_i \geq \tau]$   $\triangleright$  Eq. (1); every flow is classified
8:    $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{\hat{y}_i\}$ 
9:   if  $\hat{y}_i = \text{DDoS}$  then  $\triangleright$  positive prediction
10:     $\sigma \leftarrow \text{Signature}(f_i)$   $\triangleright$  attack-episode identity
11:    if  $\mathcal{P}.\text{Admit}(\sigma, t_i)$  then  $\triangleright B(f_i) = 1$ ; Eq. (2)
12:       $\phi_i \leftarrow \mathcal{E}.\text{SHAPValues}(\mathbf{x})$   $\triangleright$  Eq. (6)
13:       $\mathcal{P}.\text{Record}(\sigma, \phi_i)$ 
14:       $\mathbf{a}_i \leftarrow \phi_i$   $\triangleright$  exact for  $f_i$ 
15:    else
16:       $\mathbf{a}_i \leftarrow \mathcal{P}.\text{Representative}(\sigma)$   $\triangleright \phi_{\rho(i)}$ , by
        reference
17:    end if
18:     $\mathcal{A} \leftarrow \mathcal{A} \cup \{\text{FormatAlert}(f_i, \hat{y}_i, s_i, \mathbf{a}_i)\}$ 
19:  end if
20: end for
21: return  $\mathcal{Y}, \mathcal{A}$ 

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the budget, which is the component that separates the cost of classification from the cost of explanation.

Two properties of Algorithm 1 are worth drawing out, because they determine the cost figures reported later. Classification occurs on line 7 for every flow without exception, so the classification rate equals the flow rate. Exact TreeSHAP attribution occurs on line 12 only when the budget admits the alert, so the rate at which attributions are *computed* is set by policy and not by the traffic mix. Every alert still leaves with an attribution attached, but not the same kind: an admitted alert carries its own exact vector, set on line 14, while a flow whose episode has already been explained carries a reference to the representative's vector, set on line 16. The audit trail is therefore complete in the sense of Eq. (4), every alert resolving to an attribution, and incomplete in the sense that only the admitted fraction resolves to an attribution of its own score. Setting $\mathcal{P}$ to admit unconditionally recovers the unbudgeted behavior in which the two rates coincide and every alert carries an exact per-flow attribution.

A. RANDOM FOREST CLASSIFICATION

Random Forest [33] is an ensemble of T trees $\{h_t(\cdot)\}_{t=1}^T$, each fitted to a bootstrap sample with a random feature subset at

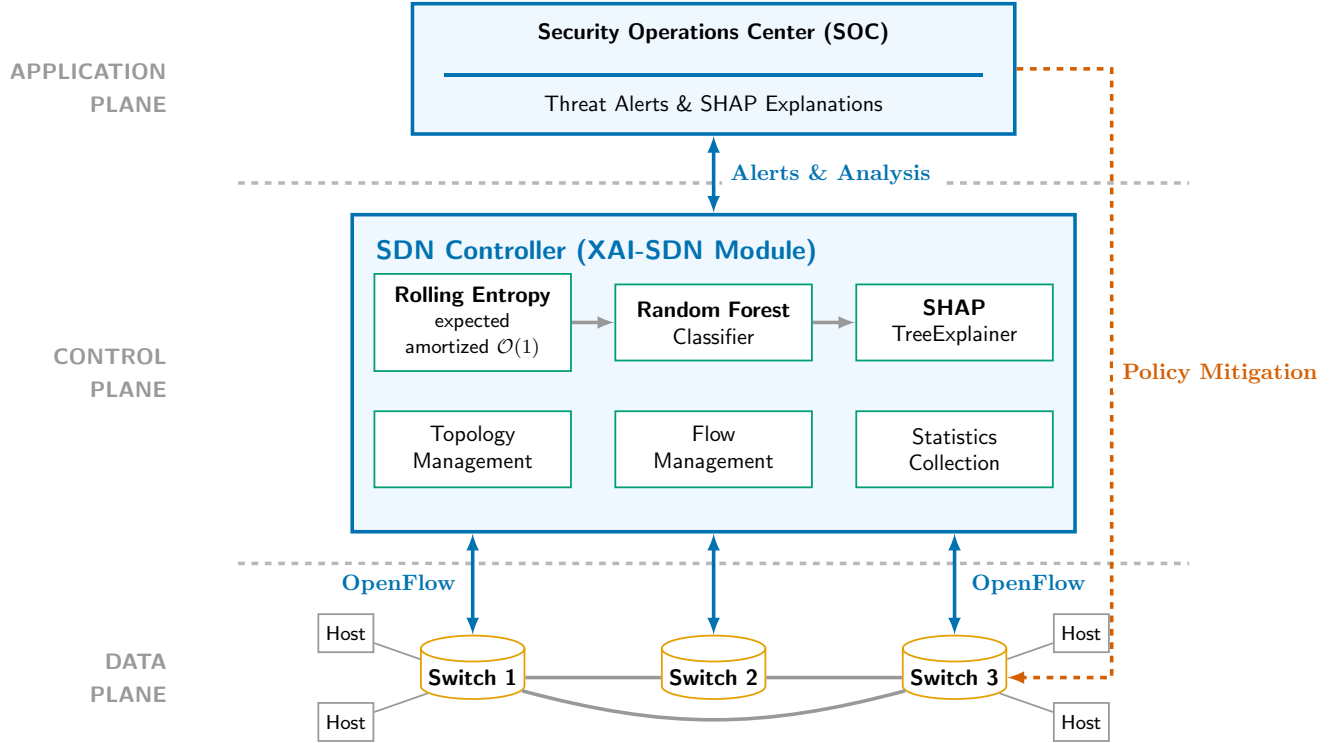


FIGURE 1. XAI-SDN system architecture. OpenFlow switches export flow statistics to the controller-embedded XAI-SDN module, which performs entropy-augmented feature extraction at expected amortized $\mathcal{O}(1)$ cost per update under the data-structure assumptions of Section V-C, Random Forest classification, and TreeSHAP attribution for the alerts the explanation budget admits. Detected DDoS events trigger alerts and policy updates; the dashed arrow represents the southbound policy-enforcement feedback loop.

each split, and the label for $\mathbf{x}$ is the majority vote:

$$\hat{y} = \arg \max_{c \in \mathcal{C}} \frac{1}{T} \sum_{t=1}^T \mathbb{I}[h_t(\mathbf{x}) = c], \quad (5)$$

where $\mathcal{C} = \{\text{Benign}, \text{DDoS}\}$ and node splits minimize the Gini impurity $G(t) = 1 - \sum_j \hat{p}_{ij}^2$. The fraction of votes for the DDoS class serves as the score $g_\theta(\mathbf{x})$ in Eq. (1).

B. SHAP EXPLAINABILITY

Subscripts are used consistently throughout: i indexes flows and j indexes the d features of a flow, so ϕ_{ij} is the contribution of feature j to the score of flow f_i and ϕ_i is the vector of all d of them. SHAP [28] obtains ϕ_{ij} by averaging the change in model output that adding feature j produces over every subset S of the remaining features, relative to the expected output. Writing $g_\theta(S)$ for the model's output when only the features in S are known,

$$\phi_{ij} = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|! (|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} [g_\theta(S \cup \{j\}) - g_\theta(S)], \quad (6)$$

where $\mathcal{F} = \{1, \dots, d\}$ is the complete feature set. These values satisfy the additive efficiency property of Eq. (3), which is what makes an attribution auditable: the score decomposes exactly across features with no residual. TreeSHAP [29] computes them exactly for tree ensembles in $\mathcal{O}(TLD^2)$ time, with L the maximum leaves and D the maximum depth per tree.

C. EVALUATION METRICS

Detection quality is reported from the confusion matrix, treating DDoS as the positive class. Writing TP, FP, TN, and FN for the four counts, Eq. (7) defines accuracy, the false positive rate, and the false negative rate:

$$\begin{aligned} \text{Acc} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \\ \text{FPR} &= \frac{\text{FP}}{\text{FP} + \text{TN}}, \quad \text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}}. \end{aligned} \quad (7)$$

We also report per-class precision P_c and recall R_c , their harmonic mean $\text{F1}_c = 2P_c R_c / (P_c + R_c)$, and macro F1, the unweighted mean of the per-class F1_c , which is the accuracy-like summary that does not collapse to the majority class under severe imbalance.

Threshold-free performance is summarized by two curves rather than one. The area under the receiver operating characteristic curve (ROC-AUC) is reported for continuity with prior work, and the average precision (AP), the area under the precision-recall curve, is reported because it is the informative quantity at the class ratio evaluated here. The distinction is not cosmetic. A classifier drawing at random attains ROC-AUC 0.5 regardless of prevalence, but attains AP equal to the positive prevalence, which on our test partition is above 0.97. ROC-AUC therefore remains a useful prevalence-independent measure of ranking discrimination,

but at this class ratio it does not characterize the precision or the false-alarm burden at the operating point actually deployed, whereas AP, precision, and the absolute count of false alarms do. For the same reason, threshold selection in Section VII is performed on the precision–recall plane rather than on the false negative rate, which is normalized by the majority class and moves negligibly as individual errors are added or removed.

V. MATHEMATICAL FOUNDATION

The entropy features are specified below in full: what each one summarizes, which exported field it is computed from, how continuous quantities are discretized, how the sliding window is maintained, what the update recurrence is, and what that recurrence costs in both time and numerical accuracy. The level of detail is deliberate. An entropy feature is only reproducible if the window semantics and the binning are stated, and its complexity claim is only meaningful if the supporting data structures are named.

A. ENTROPY FEATURE EXTRACTION

Shannon entropy [34] measures how concentrated a discrete distribution is. Applied to a window of recent flows it summarizes the diversity of an attribute: source addresses, destination ports, protocol numbers, and so on. Concentration is often what a flood produces, so low entropy over a window may indicate an attack in progress and higher entropy is commonly associated with ordinary traffic. The relationship is statistical rather than deterministic, and it depends on the attribute chosen, the window length, the composition of the background traffic, and the attacker’s strategy. We treat entropy as evidence for a classifier to weigh, not as a decision rule.

Given a sliding window W holding the N most recent flows, the entropy of source addresses is

$$H_{\text{src}}(W) = - \sum_{i=1}^{|\mathcal{I}|} \frac{n_i}{N} \log_2 \frac{n_i}{N}, \quad (8)$$

where $\mathcal{I}$ is the set of distinct source addresses present in W and n_i is the number of flows in W carrying the i -th address. Equation (8) is evaluated identically for each of the eight attributes in Table 3; only the key changes. Continuous attributes are discretized by flooring to a fixed bin width before the entropy is taken, and the bin widths are given in the table rather than left implicit, since entropy over a continuous quantity is undefined without them.

Two properties of this feature set require explicit statement. First, H_{len} and H_{iat} are computed over per-flow means, not over per-packet distributions, because the mean is what the flow exporter reports; they summarize the diversity of flow-level averages within a window and should be read that way. Second, H_{flag} summarizes the distribution of SYN flag counts rather than of a full TCP flag bitmask, for the same reason. Flow records exported by CICFlowMeter contain no TTL field, so a TTL entropy feature is constant on these records

TABLE 3. Specification of the eight entropy features: source field, transformation, and bin width.

Feature	Source field (exported)	Bin width
H_{src}	Source IP	categorical
H_{dst}	Destination IP	categorical
H_{dport}	Destination Port	integer, 1
H_{proto}	Protocol	integer, 1
H_{len}	Packet Length Mean	10 bytes
H_{iat}	Flow IAT Mean	1000 μ s
H_{flag}	SYN Flag Count	integer, 1
H_{sport}	Source Port	integer, 1

and carries no information. Source-port entropy is used in its place: it is exported, it is non-degenerate, and it is directly relevant to floods that randomize source ports. The dimensionality of the representation is $d = 88$ in either case.

B. WINDOW MAINTENANCE AND THE UPDATE RECURRENCE

The window is a fixed-capacity circular buffer of flow keys paired with a hash map from key to multiplicity. Admitting flow f_i appends its key and increments that key’s multiplicity; once the buffer is full, each admission also evicts the oldest key and decrements its multiplicity, deleting the entry when the count reaches zero. Before the buffer fills, entropy is computed over the $|\mathcal{W}| < N$ flows seen so far, so the first N decisions of a cold start are taken on a partially populated window. In the offline evaluation each partition is processed with a freshly initialized window, which prevents state from one partition influencing another.

Recomputing (8) from scratch on every admission would cost $\mathcal{O}(|\mathcal{I}|)$. Instead the engine maintains the scalar

$$S = \sum_{k \in \mathcal{K}(W)} n_k \log_2 n_k, \quad (9)$$

where $\mathcal{K}(W)$ is the set of distinct keys in W , and recovers the entropy as

$$H(W) = \log_2 |W| - \frac{S}{|W|}. \quad (10)$$

Admitting a key with current multiplicity n_a and evicting one with current multiplicity n_d updates S by

$$S' = S - \psi(n_a) + \psi(n_a+1) - \psi(n_d) + \psi(n_d-1), \quad (11)$$

with $\psi(m) \triangleq m \log_2 m$ and $\psi(0) \triangleq 0$.

Equations (9)–(11) are the recurrence the implementation executes. They are stated in this form deliberately: an update written directly on H , using terms of the form $-(m/N) \log_2 (m/N)$, is only valid while $|W|$ is constant and therefore fails during window warm-up, whereas the S -form is correct throughout because the $|W|$ dependence is factored out into (10).

C. COMPLEXITY AND NUMERICAL BEHAVIOR

Each admit/evict pair performs a constant number of buffer operations, a constant number of hash-map operations, and a constant number of arithmetic operations, independent of N and of $|\mathcal{K}(W)|$. The circular buffer gives worst-case $\mathcal{O}(1)$ append and eviction. The hash map gives *expected* $\mathcal{O}(1)$ lookup, insertion, and deletion; under adversarial key collisions the worst case degrades to $\mathcal{O}(|\mathcal{K}(W)|)$ for an individual operation. The correct statement is therefore that the engine performs *expected amortized* $\mathcal{O}(1)$ work per update under these data-structure assumptions, not unconditional $\mathcal{O}(1)$. Where periodic exact resynchronization of S is enabled, as discussed below, it contributes a further $\mathcal{O}(|\mathcal{K}(W)|/R)$ amortized per update for a resynchronization interval of R . So that the bound can be verified against a measurement, the running-sum recurrence costs **381.5 ns** per admit/evict pair on the machine of Table 5, measured over the 1.2×10^7 -update stream of Section VII-I. An alternative formulation of the same update, written directly in terms of φ , costs 722.3 ns on the same stream.

Maintaining S incrementally accumulates floating-point error, since each update adds and subtracts terms of comparable magnitude. This is a legitimate concern for a detector intended to run continuously, and Section VII reports the measured drift over more than 10^7 updates, its effect on classifier decisions, and the cost of two mitigations: compensated summation, and periodic exact recomputation of S from the multiplicity map.

D. FEATURE VECTOR

The classifier input concatenates the 80 exported flow statistics with the eight entropy features of Table 3:

$$\mathbf{x} = [f_1, \dots, f_{80}, H_{\text{src}}, H_{\text{dst}}, H_{\text{dport}}, H_{\text{proto}}, H_{\text{len}}, H_{\text{iat}}, H_{\text{flag}}, H_{\text{sport}}], \quad (12)$$

where $\{f_j\}_{j=1}^{80}$ are the CICFlowMeter statistics and $\{H_k\}$ the eight entropy metrics, giving $d = 88$. Equation (12) is the vector scored by (1) and attributed by (6).

VI. EXPERIMENTAL SETUP

Four things are fixed before any result is reported: the corpora, the computing environment, the evaluation protocol, and the hyperparameter search. We state them so that every number reported later can be traced to a specific partition, a specific machine, and a specific selection procedure. Two matters are stated explicitly because they determine how every subsequent figure should be read: how the train and test partitions are formed, and which observations were permitted to influence tuning.

A. DATASETS

The primary corpus is the SYN-flood partition of CIC-DDoS2019 [4], the 1.87 GB 03-11/Syn.csv capture. It holds 4,320,541 exported flow records. Rows carrying missing or infinite values are removed (283,076 rows), duplicate

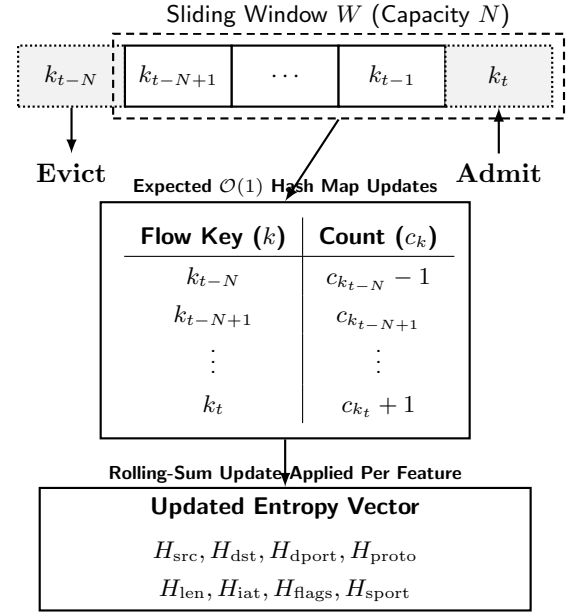


FIGURE 2. Rolling entropy engine. A circular buffer of the N most recent flow keys and a hash map of key multiplicities let each admit/evict pair update all eight entropy features with expected amortized $\mathcal{O}(1)$ work, independent of N . The update applied per feature is Eq. (11).

feature vectors are collapsed (446,671 rows), and labels are mapped to a binary target, leaving 3,590,794 flows of which 31,014 are benign. Flows are then ordered by export timestamp.

Generalization is assessed on five further corpora rather than on the SYN partition alone. Within CIC-DDoS2019 we use the LDAP, NetBIOS, and Portmap partitions of the same capture day to test transfer across attack vectors, and the SYN partition of the separate 01-12 capture day to test transfer across background conditions. Beyond CIC-DDoS2019 we use InSDN [22], collected on a real SDN testbed, and the DDoS and benign captures of CIC-IDS2017 [21]. Table 4 summarizes all of them.

Because these corpora were produced by different versions of the flow exporter, cross-corpus transfer is performed on the intersection of their schemas, computed explicitly and applied symmetrically so that the source model is trained on exactly the columns available at the target. Transfer to InSDN retains 79 flow statistics and all eight entropy features. CIC-IDS2017 exports no address, source-port, or protocol column, so four entropy features cannot be computed there and transfer retains 78 statistics and four entropy features. This is a property of the published corpora, not a modeling choice, and results are reported accordingly.

All corpora are publicly released and pre-anonymized. No additional personally identifiable information was collected or processed.

B. EVALUATION PROTOCOL

Split methodology turns out to matter more on this benchmark than any modeling decision we make, so the protocol is stated

TABLE 4. Corpora used in this study, after cleaning.

Corpus	Role	Flows	Benign	Benign (%)
CIC-DDoS2019 SYN (03-11)	primary	3,590,794	31,014	0.86
CIC-DDoS2019 LDAP	vector	1,200,000	1,291	0.11
CIC-DDoS2019 NetBIOS	vector	1,200,000	827	0.07
CIC-DDoS2019 Portmap	vector	181,860	4,695	2.58
CIC-DDoS2019 SYN (01-12)	cross-day	1,200,000	356	0.03
InSDN	cross-set	223,372	68,419	30.63
CIC-IDS2017 DDoS	cross-set	223,082	95,068	42.62

before any result.

The primary protocol is a *chronological* split, referred to below as protocol A: flows are ordered by export timestamp, the first 70% form the training partition and the last 30% the test partition, so that no training flow postdates any test flow. On the SYN partition this yields 2,513,556 training and 1,077,238 test flows, with 30,432 benign flows in the test partition, or 2.82%. Where a threshold or a hyperparameter must be selected, a validation block is held out chronologically from the end of the training partition and the selection is made there; the test partition is read exactly once, at the reported operating point. Two validation blocks of different sizes appear later and we name both here rather than leaving a reader to reconcile them. Threshold selection (Section VII-H) uses the last 10% of the *whole corpus*, which is 359,080 flows holding 60 benign, leaving 2,154,476 for fitting. Model-size selection (Section VII-Q) uses the last 10% of the *training partition*, which is 251,356 flows holding 49 benign, leaving 2,262,199 for fitting. Neither block touches the test partition.

We also report, for comparison only, a *stratified random* split of the same proportions, which assigns 30% of each class to the test partition irrespective of time. This is the protocol implicitly used by much of the literature on this benchmark. Reporting both is not redundant. Section VII shows that the difference between them accounts for more of the gap between near-perfect and realistic performance figures than any architectural choice examined in this paper, and a reader comparing our numbers against published ones needs to know which convention produced each.

A third protocol exists because the baseline comparison needs it, and it is defined here rather than in the table that uses it. Protocol B is a *subsampling* chronological protocol applied to every method in Section VIII: 100,000 training and 40,000 test flows drawn at the corpus's own class ratio rather than balanced, giving 31 benign flows in training and 1,174 in test, or 2.94%. The intent was a contiguous temporal block at each end; a contiguous block of that size is single-class on this corpus, so the draw falls back to a stratified subsample of the same sizes, and we state that rather than describing an intent the run did not execute. Protocol B exists so that models which cannot be fitted on 2.5 million flows can be compared against the rest under one procedure. It is reported in its own tables and never mixed with protocol A, because the two answer different questions: protocol A describes the setting

the detector is claimed to operate in, protocol B describes a comparison the full-scale setting cannot support. One model is capped further within protocol B: the RBF support vector machine is fitted on 40,000 of the 100,000 training flows, because its kernel computation is quadratic in the sample count, and that cap is stated wherever its figures appear.

Hyperparameters are selected by nested cross-validation with forward-chaining outer folds, so that each fold trains on an expanding temporal prefix and tests on the block that follows it. Five outer folds are run over a grid of sixteen configurations, with the inner selection on a validation block carved from the last 15% of each outer training prefix. The outer test block is never read during selection or fitting. The procedure is reported here for what it establishes about the corpus rather than for a configuration it recommends: the five folds select *five different* configurations, so hyperparameter selection is not stable on this data, and the settings of Table 5 are library defaults rather than a configuration this procedure endorses. Section VII-Q shows the same instability from the other direction, with every configuration scoring identically on the validation block.

C. COMPUTING ENVIRONMENT

Every timing figure in this paper was measured on the single machine described in Table 5, on the CPU, with the same thread budget and the same measurement harness for every method. The installed PyTorch build carries no CUDA support, so no model in the comparison can be timed on a GPU even inadvertently. Latency and throughput figures across all methods and all tables are therefore directly comparable with one another. They are not comparable with figures published elsewhere on different hardware, and we make no such comparison.

D. IMPLEMENTATION, HYPERPARAMETERS, AND REPRODUCIBILITY

XAI-SDN is implemented in Python using scikit-learn [35] and SHAP [28]; every measurement reported here was produced on the interpreter and library versions of Table 5, and each released result file carries that version block so any figure can be traced to the build that produced it. The detector uses 200 fully-grown trees with the remaining settings as listed. These are library defaults rather than the output of the selection procedure of Section VI-B. All random seeds, artifact hashes, and configuration files are archived to support reproducibility.

VII. RESULTS

This section reports detection performance under the chronological protocol, then works through the questions that performance raises. We begin with the effect of split methodology, because it turns out to dominate, and because it determines how every subsequent number should be read. We then audit the data for leakage, quantify redundancy among the features and the stability of the attributions built on them, select the operating threshold on the precision–recall plane, measure the

TABLE 5. Measurement environment and detector configuration. Every timing figure in this paper was produced on this machine, on the CPU. The tree depth is the scikit-learn default rather than a selected value; Section VII-Q reports what bounding it recovers.

Machine		Detector	
CPU	Core i9-13900H, 14C/20T	Trees T	200
Base clock	2.60 GHz	Max. depth	none (fully grown)
Cache	11.5 MB L2, 24 MB L3	Features/split	$\lfloor \sqrt{88} \rfloor = 9$
Memory	47.6 GB	Bootstrap	enabled
OS	Windows 11 Pro 26120	Class weight	balanced
GPU	none used	Window N	1,000 flows
Python	3.11.9	Threshold τ	0.70
scikit-learn	1.8.0	Seed	42
NumPy / pandas	2.3.5 / 2.3.3		
SHAP	0.51.0	Stack	
XGBoost / LightGBM	3.2.0 / 4.7.0	OpenFlow	os-ken, OF 1.3
PyTorch	2.14.0+cpu (no CUDA)	Emulator	Mininet 2.3.0, OVS 3.7.1

numerical behavior of the streaming entropy engine, test generalization across attack vectors, capture days and corpora, and finally account for the cost of running the detector inside a live control plane.

One convention applies throughout. In the tables that follow, bold marks the row the accompanying text is about, which is not in general the best row in any column: several bolded rows are beaten elsewhere in their own table, and we say so where it matters. Where bold instead marks the best value in a column, the caption states that rule explicitly.

A. SENSITIVITY TO THE EVALUATION PROTOCOL

The same model, the same features, and the same hyperparameters produce materially different headline figures depending only on how the data are partitioned (Table 6). We report this first because it determines how every subsequent number should be read.

Under a stratified random 70/30 split, the detector reaches 99.9981% accuracy and 99.9458% macro F1, with a false positive rate of zero on the test partition. Under a chronological 70/30 split of the same cleaned data, in which every training flow precedes every test flow, accuracy falls to 99.7538% and macro F1 to 97.8467%, while the false negative rate rises by more than two orders of magnitude.

The mechanism is visible in the class composition. A stratified split assigns exactly 30% of each class to the test partition, leaving 9,304 benign flows, or 0.86%. A chronological split leaves 30,432 benign flows, or 2.82%, because benign traffic in this capture is concentrated late in the recording rather than distributed evenly through it. A random split therefore disperses near-identical flows from the same short benign interval across both partitions, and the model is asked at test time about traffic it has already seen close analogues of.

We report this prominently for two reasons. It is the largest single effect we measure, larger than any architectural choice examined in this paper. It also explains a good deal of the near-perfect performance reported on this benchmark generally, and a reader comparing our figures against published ones needs to know which convention produced each. All results

TABLE 6. Effect of split methodology. Identical data, model, hyperparameters, and feature definition; only the partitioning differs. Both columns use the unrepaired feature definition, including the degenerate TTL entropy feature discussed in Section V-A, so that the comparison isolates the split and nothing else. The headline figures of Section VII-B use the repaired feature set and differ from the chronological column here by 0.03 macro F1 points.

Metric	Stratified random	Chronological
Training flows	2,513,555	2,513,556
Test flows	1,077,239	1,077,238
Benign in test	9,304	30,432
Benign in test (%)	0.86	2.82
Accuracy (%)	99.9981	99.7538
Macro F1 (%)	99.9458	97.8467
ROC-AUC	1.000000	0.999721
False positive rate (%)	0.0000	0.1117
False negative rate (%)	0.0019	0.2501

in the remainder of this paper use the chronological protocol unless stated otherwise.

B. DETECTION PERFORMANCE UNDER THE CHRONOLOGICAL PROTOCOL

Table 7 reports the operating point of the full 88-dimensional detector on the chronological test partition of 1,077,238 flows, of which 30,432 are benign. Accuracy is 99.7579%, macro F1 is 97.8813%, ROC-AUC is 0.999728, and average precision is 0.999992. The false positive rate is 0.0953% and the false negative rate is 0.2464%.

The gap between ROC-AUC and macro F1 is the point made in Section VI in concrete form. At 97.2% positive prevalence a ROC-AUC of 0.9997 still reports genuine ranking quality, but it is insensitive to the minority class the operator actually cares about, so it does not by itself characterize the false-alarm burden at the deployed threshold. Macro F1, average precision and the absolute false-alarm count are the figures that discriminate between models here, and they are what we report throughout, alongside ROC-AUC rather than in place of it. The lower half of Table 7 gives the per-class

breakdown, where the asymmetry is plain: the majority class is detected almost perfectly while the benign class carries essentially all of the error.

C. FEATURE-SET AUDIT

Two properties of the feature set are worth reporting before any ablation, because both bear on how much of the representation is doing work.

Twelve of the 80 exported flow statistics are constant across all 3,590,794 flows of this partition and therefore carry no information: `Bwd_PSH_Flags`, `Fwd_URG_Flags`, `Bwd_URG_Flags`, `FIN_Flag_Count`, `PSH_Flag_Count`, `ECE_Flag_Count`, and the six bulk-rate fields. `FIN_Flag_Count` warrants particular note: a column that never varies cannot carry attribution weight, and the attributions reported in Section VII-G do not rank it.

Ranked by direction-agnostic single-feature AUC, the eight entropy features occupy the top eight positions, ahead of every exported flow statistic. The best flow statistic, `ACK_Flag_Count`, reaches 0.8938 against 0.9986 for destination-address entropy. This probe is a descriptive statistic of the corpus and is computed over the whole partition, training and test flows together, rather than on the training partition alone. We state that because Section VII-D uses the same ranking to define which features to remove, so that removal is informed by the test partition; the direction of the resulting bias is toward removing more than a training-only ranking would, which makes the negative result there more conservative rather than less. No model in this paper is fitted or tuned on the ranking. This supports the paper's central design choice and simultaneously raises a question that Section VII-D addresses directly: a windowed statistic can separate classes either because it captures attack structure or because it encodes position within a temporally segregated capture.

D. LEAKAGE AND DATASET-ARTIFACT AUDIT

Whether performance on this benchmark reflects attack signal or dataset artifacts cannot be settled by removing a small number of features selected by inspection. The suspect set is therefore defined objectively, as every feature whose single-feature AUC reaches 0.99, and is removed in its entirety.

Five features meet that criterion, all of them entropy features: H_{dst} , H_{src} , H_{len} , H_{proto} , and H_{dport} . Table 8 reports performance as features are removed in order of separability.

Removing the whole suspect set leaves macro F1 at 97.9394%, marginally above the 97.8813% of the full model, and removing the ten most separable features raises it to 98.6911%. No small subset is load-bearing. We state the implication in both directions, because it cuts both ways: the result rules out a narrow leakage explanation, and it also shows that the discriminative information in this partition is distributed widely enough that feature-level ablation cannot by itself establish that the model has learned attack behavior.

No feature exhibits completely disjoint class ranges, which rules out the crudest form of generator artifact. After dedupli-

cation no test flow has a feature-identical twin in the training partition, and the median standardized distance from a test flow to its nearest training neighbor is 0.142, with no test flow closer than 10^{-6} .

Two controls bound the interpretation. Permuting the labels collapses the detector to chance, macro F1 0.4978 and ROC-AUC 0.5027, which is the result a pipeline with no label leakage must produce and which licenses the remaining numbers. Shuffling the row order before splitting restores macro F1 to 0.9995, reproducing the stratified figure and confirming that the gap in Table 6 is attributable to ordering alone.

Bootstrap resampling with 1,000 replicates puts the full model's macro F1 at 97.88% with a 95% interval of [97.81, 97.97] and the suspect-free model at 97.94% with [97.86, 98.02]. The intervals overlap, so removing the flagged features produces no significant change. False positive rates are 0.096% [0.065, 0.130] and 0.122% [0.085, 0.162] respectively.

1) How much of the result was available to memorization

The census above is taken after deduplication, where it is zero by construction. What matters is what those duplicates would have done had they been kept, so we repeat the analysis on the raw partition. Before cleaning, 446,671 of 4,037,465 rows are exact duplicates, a rate of 11.06%, rising to 12.36% among benign flows. Whether that matters depends entirely on the split.

Under a stratified random 70/30 split, 15.49% of test rows have a feature-identical twin in the training partition, rising to 17.40% among benign flows. A lookup table that stores the training label of each distinct feature vector and predicts the majority class for anything unseen, learning nothing whatever, reaches 99.28% accuracy on that test partition. Under the chronological split the straddle rate falls to 0.036% overall and 0.003% for benign flows, and the same lookup table reaches 97.14%, which is simply the majority-class rate.

This does not show that the submitted model was a lookup table. It shows that on the stratified protocol 99.28 of the 99.9981 percentage points of accuracy were obtainable without learning anything about attacks, so the informative margin was roughly seven tenths of a point rather than the figure the headline suggests. On the chronological protocol the memorizable component collapses and the margin the model must earn is correspondingly larger.

E. WHY THE PROTOCOLS DIFFER: THE BENIGN CLASS IS NOT STATIONARY

The gap in Table 6 has a single cause, and it is a property of the capture rather than of any modeling decision.

Timestamps parse without error: zero of 4,320,541 records fail, the two candidate day-order conventions agree on every row, and the capture spans 6.1 hours. On that verified ordering, the median benign flow sits at position 0.995 of the timeline. Ninety-seven point nine percent of benign flows fall in the final 30% of the capture, and 97.3% fall in the final 10%. Attack traffic is spread throughout.

TABLE 7. Detection performance on the chronological test partition, repaired 88-dimensional feature set, at the fixed $\tau = 0.70$ of Table 5 used throughout this paper. Section VII-H reports separately what selection on the validation block returns and why this corpus does not support it. Support is the number of test flows in each class. Per-class figures are computed from the confusion matrix of Section VII-M, and macro F1 is the unweighted mean of the two class F1 values shown.

Overall				
Accuracy				99.7579%
Macro F1				97.8813%
ROC-AUC / average precision				0.999728 / 0.999992
False positive rate / false negative rate				0.0953% / 0.2464%
Test flows (benign)				1,077,238 (30,432)
Per class				
Class	Prec. (%)	Rec. (%)	F1 (%)	Support
Benign	92.1806	99.9047	95.8873	30,432
DDoS-SYN	99.9972	99.7536	99.8753	1,046,806

TABLE 8. Progressive removal of the most separable features, chronological protocol. Removal is in descending order of single-feature AUC.

Configuration	Dim.	Acc. (%)	Macro F1 (%)	FPR (%)
Full feature set	88	99.7579	97.8813	0.0953
Minus top 1	87	99.7679	97.9648	0.1446
Minus top 3	85	99.7590	97.8899	0.1347
Minus top 5 (all suspects)	83	99.7649	97.9394	0.1216
Minus top 10	78	99.8528	98.6911	0.0920
Minus top 20	68	99.8024	98.2580	0.0756

The consequence for chronological evaluation is severe. A 70/30 chronological split of the cleaned partition leaves 582 benign flows available for training against 30,432 at test, a ratio of one to fifty-two. Moving the cut point does not repair it: measured on the raw partition before deduplication, where all five cut points can be compared on one basis, training benign flows run 527 at 50/50, 586 at 60/40, 764 at 70/30, 918 at 80/20 and 968 at 90/10, so even devoting nine tenths of the capture to training leaves under a thousand. A chronological split of this partition is therefore not a harder version of the same task. It is close to a zero-shot problem for the benign class, and the 97.88% macro F1 of Table 7 should be read as a model generalizing to a class it has seen fewer than six hundred examples of.

This also explains the entropy result of Section VII-C. A windowed statistic computed over a capture whose classes are segregated in time partly encodes position within the capture, and position within the capture is almost perfectly predictive of the label here. Section VII-F separates that effect from genuine attack structure.

We report this because it bears on every published result on this partition, not only ours. Any evaluation that splits CIC-DDoS2019 randomly distributes this narrow benign interval across both partitions and obtains a benign class that is easy by construction.

F. CONTRIBUTION OF THE ENTROPY AUGMENTATION

The entropy features are the component this work is named for, so their contribution is measured rather than asserted. The measurement does not support the claim that the augmentation improves in-distribution detection on this corpus.

Four feature configurations were fitted five times each, over the archived seeds {42, 123, 456, 789, 1024}, under the chronological protocol throughout. The *natural* configuration is the full 88-dimensional vector. The *CIC-only* configuration drops the eight entropy features. The *entropy-only* configuration keeps only those eight. The *scrambled* configuration is the control that matters most: the entropy features are computed over a random permutation of the arrival order and then returned to their chronological positions, so the features retain their marginal distributions but lose all temporal coherence, while the split and every other column stay identical. Table 10 reports the outcome.

Three findings follow. The first is unfavorable to the augmentation. The entropy augmentation does not improve in-distribution detection on this corpus. The eighty exported flow statistics alone reach macro F1 98.28%, against 97.91% for the full 88-dimensional vector, and the difference favors the smaller feature set on every one of the five seeds. With five paired observations a Wilcoxon signed-rank test cannot return a p -value below 0.0625, which is what it returns, so the effect is small, consistent in direction, and unfavorable to the augmentation. The claim that the entropy features improve in-distribution accuracy is therefore not supported by these data and is not made in this paper.

Second, destroying the temporal coherence of the entropy features changes nothing measurable: 97.89% macro F1 scrambled against 97.91% natural, with a p -value of 0.4375 and the sign of the difference varying across seeds. Taken with Section VII-D, where the same scrambling drops the eight entropy features' individual AUCs from a mean of 0.98, and 0.95 at worst, to a mean of 0.55, this locates the effect precisely. The entropy features do carry genuine arrival-order information, and that information is almost entirely redundant with what CICFlowMeter has already encoded in the flow statistics. The classifier does not need it because it already

has it.

Third, the entropy features cannot carry the task alone. Eight features give 49.31% macro F1 at a false positive rate of 99.89%, which is a model that labels almost everything an attack. That figure should not be read as a property of entropy features in general. Under the chronological protocol the training partition contains 582 benign flows, as Section VII-E reports, and no eight-dimensional representation recovers a class from that.

1) Where the augmentation does earn its place

In distribution is the one setting in which CICFlowMeter has already encoded everything a distributional summary of the same window could add, so it is the setting least likely to reward the augmentation. A summary of how a traffic attribute is spread over a window is a claim about structure that ought to survive a change of attack vector, of capture day, or of network, where a raw per-flow counter such as a byte rate need not. We therefore repeated the comparison under every shift for which this paper already has data: three held-out attack vectors, a second capture day, and the two external corpora, each with three seeds. Table 9 reports it.

The result is conditional, and the condition is legible. Where all eight entropy features can be computed, the augmentation helps under shift and the per-seed ranges do not overlap: macro F1 rises from 0.6426 to 0.8364 on NetBIOS, from 0.4212 to 0.4820 across capture days, and from 0.2383 to 0.3811 on InSDN. On the two vectors where the flow statistics alone already exceed 0.96, it adds nothing measurable. Where it hurts, it hurts badly: on CIC-IDS2017 the full vector scores 0.2699 against 0.5658 for the flow statistics alone.

That last row is the informative one rather than an inconvenience. The CIC-IDS2017 export carries no address, protocol or source-port column, so four of the eight entropy features cannot be computed there and the “full” configuration on that corpus is a half-specified distributional summary rather than the augmentation this paper proposes. InSDN, which retains all eight, records the largest relative gain of any shift. The reading we take from the pair is that a partial entropy vector is worse than none, because the classifier allocates capacity to features that no longer mean what they meant in training.

We state the claim this licenses and no more. The entropy augmentation does not improve in-distribution detection on this corpus, and Section VII-K shows it is close to free. What it buys is robustness under distribution shift, in the cases where the full feature set is computable, and we have six shift conditions rather than a sample large enough to put a confidence interval around the effect. A reader should treat this as a directional finding consistent with what a distributional feature is supposed to do, not as a demonstration that it generalizes.

What the augmentation costs is the other half of the accounting, and it is small: the rolling engine adds expected amortized $\mathcal{O}(1)$ work per flow and a bounded buffer, which Section VII-I and Section VII-K quantify. A practitioner reading this section should conclude that on CIC-DDoS2019 the

TABLE 9. Macro F1 of the full 88-dimensional vector against the 80 flow statistics alone, in distribution and under every shift, mean over three seeds. “Entropy dims” is how many of the eight entropy features the target export allows to be computed. Bold marks a difference whose per-seed ranges do not overlap.

Target	Shift	Entropy dims	Flow stats	Full
SYN 03-11	none	8	0.9899	0.9670
LDAP	vector	8	0.9649	0.9666
NetBIOS	vector	8	0.6426	0.8364
Portmap	vector	8	0.9823	0.9855
SYN 01-12	capture day	8	0.4212	0.4820
InSDN	corpus	8	0.2383	0.3811
CIC-IDS2017	corpus	4	0.5658	0.2699

TABLE 10. Contribution of the entropy augmentation over five seeds under the chronological protocol (mean $\pm$ sample standard deviation). “Scrambled” recomputes the entropy features over a permuted arrival order with the split held fixed.

Configuration	Dim.	Acc. (%)	Macro F1 (%)	FPR (%)
Entropy only	8	96.893 $\pm$ 0.018	49.315 $\pm$ 0.004	99.886
Entropy scrambled	88	99.758 $\pm$ 0.004	97.886 $\pm$ 0.035	0.109
Full (natural)	88	99.761 $\pm$ 0.006	97.909 $\pm$ 0.051	0.120
Flow statistics only	80	99.805 $\pm$ 0.012	98.281 $\pm$ 0.102	0.095

augmentation is close to free and close to useless, and that we have not established a setting in which it pays. Whether it pays where the exported statistics are poorer, on a corpus whose classes are not segregated in time, or under an adversary shaping per-window distributions, are the three questions this result raises and none of them are answered here.

G. COLLINEARITY, REDUNDANCY, AND ATTRIBUTION STABILITY

Two properties of the representation bear directly on how the attributions should be used: the degree of collinearity among the features, and the stability of the attributions computed over them. Both are reported here, including where they constrain the interpretation.

The representation is severely collinear. Of 3,828 feature pairs, 61 have $|r| \geq 0.95$ and 19 have $|r| \geq 0.99$. Of the 76 non-constant features, **55 have a variance inflation factor above 10 and 32 above 100**, with a median VIF of 51.4, and the design matrix has a condition number of 3.9×10^{16} . By any conventional diagnostic the feature set is far past the point where individual coefficients would be interpretable in a linear model. Tree ensembles are not linear models and do not fail in that way, but the redundancy is real and it is what makes the removal results of Table 8 unsurprising in hindsight.

Reducing the feature set by correlation clustering costs nothing and returns latency (Table 11). Cutting at $1 - |\rho| = 0.05$ leaves 50 features and *raises* macro F1 to 97.99% while halving per-flow inference cost. Even 41 features hold 97.95%. Only at 29 features does performance fall below the full set.

TABLE 11. Feature-set reduction by correlation clustering. Batch inference latency is per flow. Bold marks the best value in each column rather than a row under discussion.

Feature set	Dim.	Acc. (%)	Macro F1 (%)	Lat. (ms)
Full	88	99.7579	97.8813	0.00219
Cluster cut 0.05	50	99.7706	97.9879	0.00113
Cluster cut 0.10	45	99.7670	97.9568	0.00136
Cluster cut 0.20	41	99.7660	97.9469	0.00154
Cluster cut 0.40	29	99.7198	97.5342	0.00145

Attribution stability is better than the collinearity would suggest, but only at the top of the ranking. Across ten seeds the five highest-attributed features are identical every time (top-5 Jaccard 1.00), and the full attribution vectors correlate at Spearman 0.956 on average, never below 0.932. Beyond the top five the ranking moves: top-10 Jaccard averages 0.710 and falls as low as 0.538. Across ten bootstrap resamples of the training data the picture is similar but weaker, with top-5 Jaccard 0.767 and Spearman 0.940.

The operational reading is specific. An operator shown the top five contributing features is looking at a stable explanation. An analyst who reads significance into the difference between the eighth and twelfth ranked feature is reading noise. The limit is stated rather than the headline, because feature importance inferred beyond the top of the ranking is not reproducible across seeds on this corpus.

H. THRESHOLD SELECTION ON THE PRECISION-RECALL PLANE

Selecting τ from a sweep reporting “zero FNR below $\tau = 0.5$ ” is uninformative under severe class imbalance; the precision–recall curve is the appropriate instrument at this class ratio. Adopting it exposes a further property of this corpus.

Carving a validation block chronologically from the end of the training partition, as Section VI-B specifies, yields 2,154,476 training flows containing 522 benign, and 359,080 validation flows containing **60 benign**. Sixty negatives cannot support a stable threshold estimate: the validation average precision is 1.000000 and the validation ROC-AUC is 1.000000, both saturated, because the positive prevalence in that block is 0.9998. The selection procedure is sound and the data cannot support it, which is a direct consequence of the temporal structure documented in Section VII-E.

We report what the procedure returns and label it accordingly. Maximizing validation F1 on the fine-grained curve selects $\tau^* = 0.940$. At that operating point the chronological test partition gives precision 1.000000, recall 0.996330, zero false alarms out of 30,432 benign flows, and 3,842 missed attacks out of 1,046,806. Test average precision is 0.999990 against a chance baseline of 0.9717, and test ROC-AUC is 0.999665.

The threshold is therefore not tuned in any meaningful sense on this corpus, and no such claim is made. The analysis

does establish a methodological point: at this prevalence the false negative rate moves by 10^{-6} per misclassified flow and conveys almost nothing, whereas precision and the absolute false-alarm count are normalized by the 30,432 negatives an operator would actually see. All threshold reporting in this paper uses the latter.

I. NUMERICAL STABILITY OF THE STREAMING ENTROPY ENGINE

An incremental update that repeatedly adds and subtracts terms of similar magnitude will accumulate floating-point error, and over a continuously running detector that error has no natural bound. The objection is well posed, so we measured it rather than argued about it.

A stream of 12,000,000 admit/evict updates over 65,524 distinct keys was driven through four implementations of the recurrence, with entropy recomputed exactly from the multiplicity map at 240 checkpoints. A rational-arithmetic evaluation at the midpoint confirms the float64 reference itself is exact to the printed precision, with an absolute difference of 0.

Worst-case drift across every variant is 2.3×10^{-13} bits. To convert that into the quantity that matters, all eight entropy features of all 1,077,238 test flows were perturbed by that magnitude and the detector rerun: **zero decisions changed**. Catastrophic cancellation is real in principle and, at this scale and in double precision, operationally absent.

Two incidental results are worth recording. Compensated summation and periodic resynchronization buy no accuracy here, so neither is enabled by default. And the φ -form of the update is both three times less accurate and nearly twice as slow as the running-sum recurrence the implementation executes, which is why the latter is the form specified in Section V-B.

J. TRANSFER: OTHER VECTORS, ANOTHER DAY, ANOTHER CORPUS

Training on one attack vector and testing on another is the first real test of whether the detector has learned attack structure or partition-specific detail. Table 12 reports three distances from the training distribution. The results comprise one positive finding and two negative ones.

Across attack vectors, transfer is strong. Trained only on SYN flooding and tested zero-shot on LDAP, NetBIOS and Portmap reflection attacks it has never seen, the detector reaches macro F1 0.969, 0.957 and 0.990, against 0.976 on its own held-out SYN traffic. Leave-one-vector-out gives the same picture from the other direction, at 0.966, 0.973 and 0.971 for the three reflection vectors. Transfer onto SYN from the other three is weaker, 0.697 at a false-negative rate of 6.9%. The representation carries across volumetric vectors, which an evaluation confined to SYN flooding cannot establish.

Across capture days it does not. Trained on the 03-11 SYN partition and tested on 01-12, macro F1 falls to 0.500. The false-negative rate stays at 3.5%, so the collapse is driven by

the benign class, of which the 01-12 partition contains only 356 flows. A macro average over a class with 356 members is not a stable quantity, and we report the figure with that qualification rather than either suppressing it or treating it as a clean negative.

Across corpora it fails outright. Confidence intervals on the error rates are reported alongside the point estimates below rather than only in the released results. On InSDN the detector reaches macro F1 0.603 and misses 57% of attacks: over 1,000 bootstrap resamples the false positive rate is 0.102% with a 95% interval of [0.078, 0.128] and the false negative rate is 57.23% with [56.98, 57.47]. On CIC-IDS2017 it reaches 0.279 and misses 98%, with a false positive rate of 17.37% [17.14, 17.61] and a false negative rate of 98.13% [98.06, 98.21], which is close to a detector that has stopped firing; the low false positive rate in the InSDN row is not a virtue, since a model that rarely alerts has few false alerts. The intervals are narrow because the test partitions hold over 200,000 flows each, so the transfer failure is not a sampling artifact. Because the corpora come from different exporter versions, each condition runs on the intersection of their schemas, computed explicitly and applied symmetrically: InSDN retains all eight entropy features, while CIC-IDS2017 carries no address, protocol or source-port column and retains four. The targets themselves are not hard. Trained on them directly the same architecture reaches macro F1 0.999 on CIC-IDS2017 and 1.000 on InSDN, and joint training recovers the same figures, so the features and the model are adequate for both. What does not survive is the transfer.

Two caveats belong with these numbers. The InSDN export available to us is already binarized and mixes several attack families, so transfer to it measures general attack detection rather than DDoS detection, and part of the 57% miss rate is attributable to attack classes the detector was never trained on. Applying to InSDN the skepticism we applied to our own corpus in Section VII-D, an in-distribution macro F1 of exactly 1.000 after deduplication is implausibly perfect and most likely means InSDN is trivially separable under a random split. That corpus has not been audited to the depth applied to our own, and the row is presented as evidence of feasibility only.

In summary, a detector trained on one corpus should not be expected to operate on another without target data. Section VII-J1 measures how much target data it takes, and Section IX-D restates the limitation.

1) What it costs to make the detector work on another network

Reporting a transfer failure and stopping leaves an operator with nothing to act on, so we measured the price of fixing it. Four conditions were compared at seven labeling budgets: the source model used zero-shot, a model trained on the target budget alone, the source forest warm-started and extended with target-fitted trees, and joint training on source data plus the budget. The train and test split of each target corpus stays chronological. Within the training partition the budget

is drawn stratified rather than as a contiguous prefix, because on both corpora the earliest training flows are single-class, and because an operator labeling their own traffic samples across it rather than taking the first n records in timestamp order. Every condition here is evaluated on the held-out 30% of the target corpus rather than on all of it, and the source model is fitted on 600,000 flows rather than the full partition, so the zero-shot figures in this subsection are not the same numbers as in Table 12 and should not be read across. The comparison that matters is between conditions within the subsection, which share a test set.

On CIC-IDS2017 the answer is encouraging and it is not flattering to transfer learning. One hundred labeled target flows, of which 35 are benign, take macro F1 from 0.3547 zero-shot to 0.9964, and five hundred take it to 0.9990. But the model that achieves this is trained on the target budget *alone*. Joint training matches it and warm-starting from the source forest is actively harmful, never exceeding 0.4613 at any budget and holding a false-negative rate above 0.91 throughout. The source model contributes nothing here, and an operator is better off discarding it and labeling a hundred flows than adapting it. We report that because it is what we measured, and because the opposite claim is the one a paper proposing a detector would prefer to make.

On InSDN no budget we tried is sufficient. The best result is 0.8513, from warm-starting at ten thousand flows, and the curve is not monotone: the same condition gives 0.7676 at five hundred flows and 0.7707 at fifty thousand. The reason is visible in the composition rather than the size of the budget. InSDN's chronological training partition holds 1,499 benign flows out of 156,360, so a stratified draw of fifty thousand contains 479 benign examples and a draw of a hundred contains five. What an operator has to supply is not n labeled flows but n labeled examples of the minority class, and on that corpus they are not available at any budget. This is the same non-stationary benign class that Section VII-E documents for our own partition, and it is a property of how these captures are built rather than of the detector.

K. THE COST OF EXPLANATION, AND HOW TO BOUND IT

Algorithm 1 triggers an attribution for nearly every flow when almost all traffic is hostile, and at that rate attribution becomes the controller's bottleneck. This section measures the resulting throughput collapse and then measures three mechanisms that bound it.

The two stages differ by a factor of fifty. Over a 60,000-flow stream at 99.998% attack prevalence, classification costs 0.028 ms per flow, sustaining 35,715 flows/s, while one Tree-SHAP attribution costs 1.485 ms, sustaining 673 explanations per second. The alert rate on this stream is 0.99997, so the unbudgeted policy explains essentially every flow and the sustained rate falls to **661 flows/s**, a factor of 54 below the classification rate. Table 13 reports what each budget policy recovers.

The mechanism that works is not a faster explainer but a trigger that distinguishes an alert from an episode. Struc-

TABLE 12. Transfer at three distances from the training distribution: other attack vectors in the same capture, a second capture day, and two independent corpora. The source model is trained on the CIC-DDoS2019 03-11 SYN partition throughout. Bracketed dimensions are the shared schema width, which differs per corpus because not every entropy feature can be computed from every export. Every row's false positive and false negative rate carries a 95% bootstrap interval, over 500 resamples for the same-corpus and cross-day rows and 1,000 for the two external corpora; the external-corpus intervals are quoted in the text and the full set is in the released results. Leave-one-out results for the three reflection vectors, and the in-distribution and joint-training references for the two external corpora, are given in the text and in the released results.

Condition	Target	Macro F1	FPR	FNR
<i>Same capture, held-out attack vector</i>				
in-distribution	SYN 03-11	0.9759	0.0092	0.0075
zero-shot	LDAP	0.9692	0.0674	0.0001
zero-shot	NetBIOS	0.9566	0.0060	0.0001
zero-shot	Portmap	0.9896	0.0077	0.0009
leave-one-out	SYN 03-11	0.6966	0.0002	0.0693
<i>Second capture day</i>				
cross-day	SYN 01-12	0.4995	0.0112	0.0347
<i>Independent corpus, zero-shot</i>				
zero-shot	InSDN (87-dim)	0.6026	0.0010	0.5723
zero-shot	CIC-IDS2017 (82-dim)	0.2788	0.1737	0.9813

turally identical flood flows are one attack, and the alert record for each can carry a reference to the episode representative, $\phi_{p(i)}$ in Eq. (4), rather than an attribution computed for itself, so every alert still resolves to an attribution while the number of TreeSHAP evaluations falls. How much it falls depends entirely on how an episode is defined, which is the finding we did not expect. A signature that includes per-flow volume terms fragments the flood into 53,817 distinct episodes and deduplicates almost nothing, giving a $1.1\times$ reduction. A signature on the target address and protocol collapses the same stream to two episodes, a reduction of 3.0×10^4 , and restores **35,652 flows/s**, which is the classification rate to within 0.2%.

The cost of that reduction is stated rather than hidden, because it is not free. A coarse episode explains why a flood against a given service is being blocked. It does not explain why one particular flow within the flood was scored the way it was, and an operator investigating a suspected false positive needs the latter. The policy is therefore a scale control, to be tightened as alert volume falls, and the middle rows of Table 13 are where a deployment that wants per-target rather than per-service explanations would sit. The token-bucket policy makes the opposite trade, keeping full per-flow fidelity for a fixed number of alerts per second and none for the rest: at 50 explanations per second it sustains 33,273 flows/s with 0.14% of alerts individually attributed. Asynchronous off-path attribution is the least useful of the three here, because a queue large enough to avoid drops simply moves the bottleneck without relieving it, sustaining 669 flows/s at a 1,024-entry queue, while a 128-entry queue reaches 1,315 flows/s only by dropping 51.2% of the attributions.

TABLE 13. Explanation budget policies over a 60,000-flow stream at 99.998% attack prevalence, all rows from one run of the same experiment. "Explanations" counts exact TreeSHAP evaluations, and "Coverage" is that count divided by the alerts eligible under the policy, so it is the fraction of alerts receiving an exact attribution of their own score, $a_i = \phi_i$ in Eq. (4). Under episode deduplication every remaining alert carries its representative's attribution $\phi_{p(i)}$ by reference: those alerts resolve to an attribution and are not left unexplained, but that attribution is exact for the representative rather than for them. * the P3 rows are measured on a 20,000-flow prefix of the same stream, so their coverage denominator is 20,000 alerts rather than 59,998.

Policy	Explanations	Coverage	Flows/s
Classification only (no attribution)	0	n/a	35,715
P0 attribute every alert	59,998	1.000	661
P1 episode, per-flow signature	53,817	0.897	735
P1 episode, per-target signature	38,046	0.634	1,031
P1 episode, per-service signature	2	0.000	35,652
P2 token bucket, 50/s	83	0.001	33,273
P2 token bucket, 500/s	839	0.014	20,505
P3 asynchronous, 128-entry queue*	9,760	0.488	1,315
P3 asynchronous, 1,024-entry queue*	20,000	1.000	669

L. GLOBAL FEATURE ATTRIBUTION

Averaging the absolute TreeSHAP value of each feature over 3,000 test flows and across ten seeds gives the ranking shown in Fig. 3. Three observations follow.

The single most influential feature is the ACK flag count, not a volumetric statistic. Source and destination address entropy follow immediately, and together those three account for most of the attributed signal. *Flow_Bytes/s* and *Destination_Port* appear lower in the ranking, at positions six and seven. *FIN_Flag_Count* does not appear at all: it is constant across every flow in this partition (Section VII-C) and therefore cannot contribute to any prediction.

Three of the top twelve features are entropy metrics. The entropy features are influential, and the two address entropies sit at ranks two and three, but they do not dominate the attribution in the way the single-feature separability reported in Section VII-C might suggest. The reconciliation is that the classifier has many correlated routes to the same information, so no individual feature receives a large share of the credit.

Only the top five entries are reproducible across seeds (Section VII-G). Ranks below that shift between runs, because from rank six downward the seed-to-seed standard deviation of the mean absolute attribution is comparable to the gaps between adjacent features. We therefore draw conclusions from the top of the ranking only, and the full ranking with its per-seed deviations is in the released results.

Fig. 3 illustrates the same attributions per flow, and provides an unplanned demonstration of that caveat. Its sample over-represents benign flows so that both classes are visible, and under that sample the leading three features are the same three, in a different order: source-address entropy first at 0.0727 mean absolute attribution, the ACK flag count second at 0.0638, destination-address entropy third at 0.0629. Membership of the top group is stable and the ordering within it is not, which is exactly what Section VII-G measures and what

TABLE 14. Four alerts drawn from actual test predictions, each shown with the three features whose signed TreeSHAP contributions are largest in magnitude, and the value each feature took. Scores are the forest vote fraction; the threshold is 0.70. The full ten-feature attribution for each case, including the positive contributions that the borderline and false-negative cases carry further down their rankings, is in the released results.

Case	Score	Three largest contributions
True positive	1.000	$H_{src}=0.00$ (+0.088); $H_{dst}=0.00$ (+0.080); ACK count=1 (+0.047)
False positive	0.780	ACK count=0 (−0.083); dst port=53 (−0.061); bytes/s= 5.8×10^7 (+0.056)
False negative	0.125	ACK count=0 (−0.091); $H_{src}=2.74$ (−0.072); $H_{dst}=4.26$ (−0.064)
Borderline	0.700	ACK count=0 (−0.050); $H_{src}=3.59$ (−0.042); $H_{dst}=4.68$ (−0.037)

an operator reading an attribution should be told.

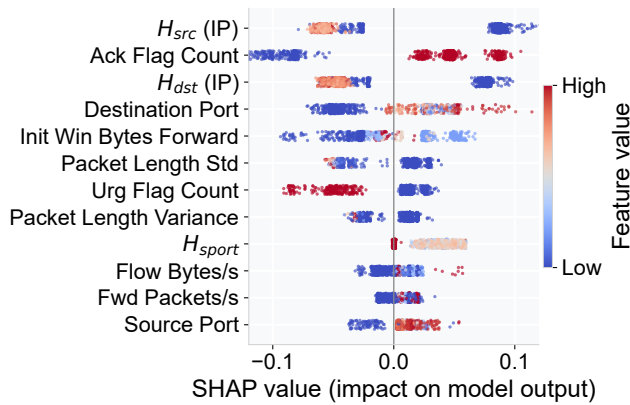


FIGURE 3. Per-flow TreeSHAP attributions for the twelve highest-ranked features, 1,500 test flows under the chronological protocol. Each point is one flow, positioned by its signed contribution to the DDoS score and colored by that feature's value. Benign flows are over-represented in this sample, at 35% rather than the 2.8% of the test partition, because a sample drawn at the natural ratio shows one class only; no number in the text is computed from this sample.

M. INDIVIDUAL ALERT CASE STUDIES

Aggregate attribution statistics say which features matter on average. They do not show whether an explanation is useful to the analyst who has to adjudicate a single alert, which is the claim the mechanism has to earn. Table 14 therefore reports four flows drawn from actual test predictions: the highest-scoring true positive, the highest-scoring false positive, the lowest-scoring false negative, and the flow whose score sits closest to the threshold. On the chronological test partition the detector produces 1,044,227 true positives, 30,403 true negatives, **29 false positives** and 2,579 false negatives, so the false positive shown is one of a very small population.

Read together the four cases describe one coherent mechanism. The true positive carries source and destination entropy of exactly zero, meaning every flow in the window shares one address pair, and those two features supply the largest positive contributions. The false negative is its mirror image:

an attack flow whose window happens to show source entropy of 2.74 and destination entropy of 4.26, and both features push the score down hard enough to keep it below the threshold. The borderline case sits between them with the same features pulling in the same direction but less strongly.

The false positive is the case that justifies the expense of per-decision attribution. It is a port 53 flow whose exported byte rate is 5.8×10^7 bytes per second, and the attribution shows the model was not persuaded by the entropy evidence at all. Both the ACK flag count and the destination port push the score down. What carried it over the threshold was the byte rate. An analyst reading that attribution can dismiss the alert on the grounds that a high-rate DNS flow is expected on this network, and can point at the specific feature responsible if the rule needs tuning. Without it the analyst sees only a score of 0.78 and has nothing to act on.

N. ROC AND PRECISION-RECALL CURVES

Fig. 4 presents ROC curves for the baselines under the full chronological protocol, paired with precision-recall curves for the same models. The pairing is substantive rather than conventional.

At 97.17% positive prevalence a classifier that predicts at random attains average precision equal to that prevalence. The majority baseline in the figure therefore sits at AP 0.9717 while its ROC-AUC is 0.5000, and the gap between a model's two figures measures how much the class imbalance flattens its ROC curve. Two rows illustrate this. The decision tree reaches AP 0.9777 against a chance value of 0.9717, an improvement of six thousandths, while its ROC-AUC of 0.6075 suggests appreciably stronger discrimination than the precision-recall figure supports. Logistic regression is worse still: AP 0.9731 is within 0.0014 of chance. Neither model is usable, and only the precision-recall panel says so.

The detector reaches ROC-AUC 0.999728 with a bootstrap interval of [0.999711, 0.999744] over 150 resamples, and average precision 0.999992. Its nearest competitor on ranking quality is LightGBM at 0.999535, and the two are separated far more clearly by their false-alarm counts than by either curve.

Curves rank; they do not choose an operating point, and the two can diverge badly. We therefore also report each model at a threshold selected on the first half of its test scores and evaluated on the second, alongside the fixed $\tau = 0.70$ used throughout the paper. For the tree ensembles the two agree to within two points. For the graph model of Section VIII-B they do not: its ROC-AUC is 0.9973, yet at any threshold selected this way it classifies almost nothing as an attack, because its outputs are compressed into a narrow band well below τ . Reporting only a curve would present that model as competitive with ours. It is competitive at ranking and not usable as a detector at any threshold selected by this procedure, and the distinction is operationally material.

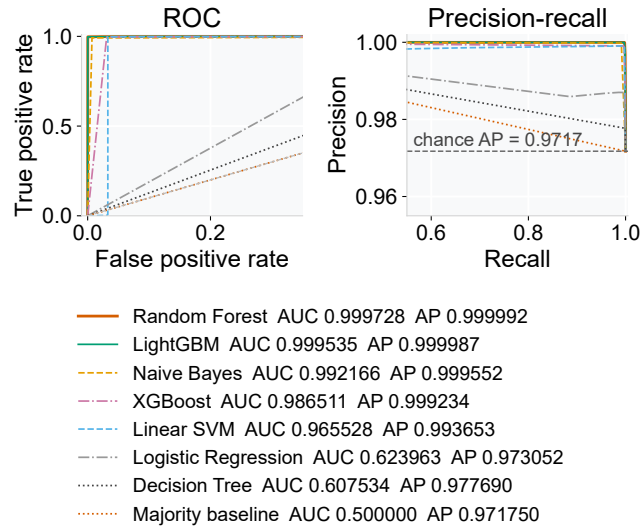


FIGURE 4. ROC and precision-recall curves on the full chronological test partition, 1,077,238 flows at 97.17% positive prevalence, every model measured by us on the same data. Axes are zoomed to the region where the curves differ. The majority baseline gives the chance operating point in both panels: ROC-AUC 0.5000 and average precision 0.9717. Legend entries are ordered by ROC-AUC.

O. HYPERPARAMETER SENSITIVITY

Two hyperparameters govern the detector: the detection threshold and the entropy window length. They are treated differently here, and the difference matters for the tuning and testing separation that Section VI-B describes.

The threshold is selected, and Section VII-H reports that selection in full, on the validation partition alone, together with the reason this corpus does not support it. We do not repeat the sweep here.

The window length was not tuned. It was fixed at $N = 1000$ flows before any evaluation and held there for every experiment in this paper. That is a deliberate choice rather than an omission. With only 60 benign flows in the validation block, a window length selected on that block would be fitted to noise, and selecting it on the test partition would invalidate every number that follows. A sensitivity sweep over N under the chronological protocol has not been performed. A sweep measured under the stratified split would not be comparable with any figure reported here, and none is carried over.

P. LIVE DEPLOYMENT ON AN EMULATED NETWORK

The testbed of Section VII-R is a control plane without a data plane: it measures what the controller costs, but no packet ever traverses a switch. The emulated network reported here answers a different question, namely whether the mitigation the detector installs protects the traffic it is intended to protect.

The network is emulated with Mininet 2.3.0 over Open vSwitch 3.7.1 (Table 5). The topology is one Open vSwitch datapath speaking OpenFlow 1.3 to the detector running inside an os-ken controller process, with three hosts on 100 Mbit links at 1 ms delay: an attacker, a benign client, and a server. The benign client runs an iperf3 transfer and a

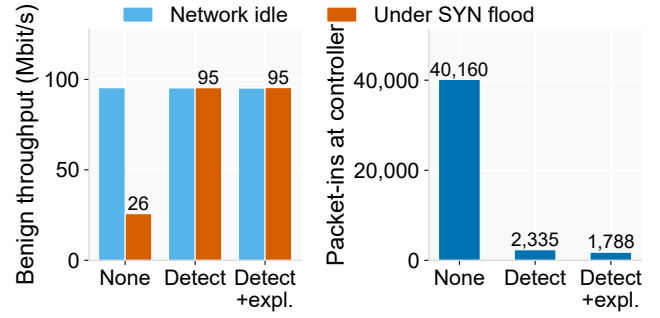


FIGURE 5. The emulated network under a randomized-source SYN flood. Left: benign iperf3 throughput with the network idle and during the flood, for each of the three conditions. Right: packet-in events reaching the controller over the same run. The conditions are abbreviated on the axes: “None” is the learning-switch controller with detection disabled, “Detect” adds the detector, and “Detect +expl.” additionally attributes every admitted alert; Table 15 names them in full. Installing drop rules at the switch both preserves the benign transfer and removes most of the control-plane load the flood would otherwise impose.

ping series against the server, first with the network idle and then while the attacker runs a randomized-source SYN flood from hping3. Three conditions are compared. A recording of one such session is submitted as supplementary material. Detection is applied to TCP only, because the detector is trained on TCP flow records and scoring an ICMP packet would evaluate the model on a vector whose TCP fields are all zero.

Table 15 reports the result and Fig. 5 shows the two comparisons that carry it. The first row is the one that matters. With no detector the flood costs the benign client 73% of its throughput, falling from 95.39 to 25.78 Mbit/s. With the detector the same flood costs it nothing measurable: 95.29 to 95.39 Mbit/s, zero added packet loss, and a ping round-trip time that does not rise. The switch holds 997 drop rules against distinct spoofed sources by the end of the attack, against four flow entries and no drop rules in the undefended condition.

Two further observations come out of the same run. Mitigation pays for its own control-plane cost: packet-in events reaching the controller fall from 40,160 undefended to 2,335 with detection enabled, because the installed rules stop the flood at the switch, so the controller stops seeing it. And attributing every admitted alert costs 21% more decision latency, 10.84 ms against 13.12 ms at the median, and 34% more resident memory, with no measurable difference in delivered quality of service. That is the same trade Section VII-K quantifies offline, now visible on a data path.

This is a three-host emulation and we do not present it as more. It cannot exhibit multi-datapath rule distribution, controller failover, or the traffic patterns of an operational fabric, and Section IX-D says so. What it does establish is that the mitigation path works end to end against live traffic, which no offline measurement in this paper can.

TABLE 15. Emulated network under a randomized-source SYN flood. Throughput is a 100 Mbit/s iperf3 transfer by the benign client; the flow table is read at the end of the attack. The undefended row uses the same learning-switch controller with detection disabled, so the comparison isolates the detector rather than the presence of a controller.

	No detector	Detect	Detect + explain
Throughput idle (Mbit/s)	95.39	95.29	95.20
Throughput under attack	25.78	95.39	95.48
Fraction retained	0.270	1.001	1.003
Added packet loss (%)	0.0	0.0	0.0
Flow entries / drop rules	4 / 0	999 / 997	621 / 619
Packet-ins at controller	40,160	2,335	1,788
Controller CPU (% of a core)	32.5	101.0	100.9
Controller memory (MB)	58	190	255
Decision latency p50 (ms)	n/a	10.84	13.12

Q. MODEL SIZE, LATENCY AND WHAT THE CONFIGURATION ACTUALLY COSTS

The detector's configuration was not selected: 200 trees grown to unbounded depth are the library defaults. With a single-flow decision costing 7.05 ms against 0.26 ms for LightGBM at the same macro F1, a natural hypothesis is that the trees are large and that bounding their depth would recover most of the gap. Depth and ensemble size were swept jointly to test it. The hypothesis is not supported, and the manner of its failure identifies the actual cost driver.

The trees are not enormous. Across the whole grid, including unbounded depth, they average between 32.7 and 46.1 leaves, and the largest forest in the grid serializes to 1.47 MB. The cause is the class balance rather than the depth limit: with 533 benign flows in a 2,262,199-flow training partition and class-balanced weighting, splitting stops early for want of minority examples, so an unbounded tree and a depth-8 tree end up nearly the same size. A reader who assumed, as we did, that the default configuration was pathological would have drawn the wrong conclusion from the latency figure.

Detection quality is equally insensitive. Table 16 reports the frontier, one row per ensemble size at three depths. Macro F1 across all 24 configurations spans 0.9778 to 0.9799 and the false positive rate spans 0.09% to 0.14%. Neither depth nor ensemble size is load-bearing on this corpus, which is consistent with Section VII-D: the discriminative information is spread widely enough that the model has many routes to it.

What does scale is memory, and with it latency. The ensemble size sets the serialized footprint almost linearly, from 0.14 MB at 25 trees to 1.47 MB at 200. Taking the smallest configuration whose macro F1 is within 0.005 of the best gives depth 8 with 25 trees: a tenfold smaller model for a macro F1 change of -0.0009 and a false positive rate 0.02 percentage points higher. On a controller, or on the constrained gateway of Section IX-B, that is worth taking on memory grounds alone, and it turns out to buy latency too.

Latency needs a measurement the sweep cannot give. Each configuration in the sweep is timed immediately after being fitted, while the fitting thread pool is still winding down, and

TABLE 16. Model-size frontier, chronological protocol, evaluated on all 1,077,238 test flows. One row per ensemble size at three depths; the full grid of six depths is in the released results. Macro F1 spans 0.002 across every configuration measured.

Depth	Trees	Size (MB)	Leaves/tree	Macro F1	FPR (%)
8	25	0.14	35.2	0.9782	0.12
8	200	1.06	32.7	0.9778	0.10
16	25	0.18	44.8	0.9799	0.14
16	200	1.44	45.0	0.9790	0.10
unbounded	25	0.18	45.3	0.9799	0.12
unbounded	200	1.46	46.0	0.9791	0.10

the resulting figures are non-monotonic in model size and differ by a factor of two between neighboring configurations. We therefore fitted only the two endpoint configurations in a single process and probed them alternately over five rounds of 2,000 calls each, so that any machine state affects both equally.

Pooling all 10,000 probes per configuration gives a median of 6.61 ms for the inherited configuration against 0.95 ms for the selected one: a $7.0\times$ reduction for the -0.0009 macro F1 already noted. Latency is therefore close to linear in ensemble size, at 0.033 ms per tree for the 200-tree forest and 0.038 ms per tree for the 25-tree one, and almost independent of the depth limit, which is what the uniform 46-leaf tree size predicts. The order-of-magnitude gap against LightGBM in Table 19 is a per-tree property of the prediction path rather than anything a change of ensemble size closes. Section VII-K reports the batched rate of 35,715 flows/s, which is what the detector achieves when the per-call cost is amortized across a batch.

One caveat applies to the selection. It was meant to be made on the validation block of Section VI-B, but that block holds 49 benign flows out of 251,356 and every configuration scores identically on it, so validation cannot separate them. The choice above is therefore a reading of the frontier rather than a protocol-clean selection, and we label it as one. The corpus does not support model selection here for the same reason it does not support threshold selection (Section VII-H).

R. CONTROL-PLANE COST AND THE RATE A CONTROLLER CAN INGEST

Every throughput figure so far is an offline one: features are read from a matrix and scored in a loop. Whether 60,000 novel flows per second can be ingested without saturating the OpenFlow control channel cannot be determined from an offline measurement. We therefore run the detector inside a controller process that exchanges genuine OpenFlow 1.3 messages with switch agents over TCP, with the encodings validated against the os-ken reference parser, and measure the controller itself rather than the model. Each session binds its own port, so a controller left behind by an earlier session cannot absorb the next one's traffic.

One quantity carries this subsection and it needs a definition before any number is given. A switch agent is syn-

chronous: it writes a packet-in and blocks until the controller answers. The *achieved rate* is therefore the number of packet-in exchanges the agent completed divided by the wall time of the session, which is the rate at which the control plane actually absorbed novel flows including classification and rule installation. Two session configurations produce it, and they are not interchangeable. The *fixed-load* session offers 3,000 events per second for 150 seconds with a second agent carrying benign background traffic through the same controller. Each *sweep* session offers one rate for 45 seconds with no background agent and a freshly started controller. Every rate reported below names which of the two produced it.

Under the fixed-load session the controller absorbs 114 events per second with detection enabled, at 98.3% of one core and 176 MB resident, with a decision taking 6.61 ms at the median and 32.70 ms at the 99th percentile. Turning on attribution for every alert halves that to 61 events per second, nearly doubles the median decision to 11.33 ms, and raises resident memory to 234 MB. Benign background traffic through the same controller pays for this: its 95th-percentile round-trip time rises from 34.11 ms to 41.16 ms, an inflation of $1.21\times$. That is the quality-of-service cost of the security function, measured on unrelated flows rather than assumed to be negligible.

The sweep sessions answer a different question, and they answer it partly in the negative (Table 17). Their highest achieved rate is 77 events per second, at the lowest rate offered. That rate already returns 15.5% of what is offered, so the controller is saturated at every point of the sweep and the sweep does not locate the knee, which lies below 500 offered and which we did not measure. What it does establish is that goodput does not rise with offered load: from 1,000 offered upward it sits at 34 to 36 events per second and stays there while the median round-trip time nearly triples, so offering more work returns less, which is queue behavior past saturation rather than a gradual degradation an operator would notice from a throughput number alone.

Both are reported rather than reconciled, because they do not reconcile at the precision a single number implies. A 45-second sweep session and a 150-second fixed-load session differ in duration, in the presence of background traffic, and in how much of each session is spent on controller start-up, and the spread across sessions, 34 to 114 events per second, is wider than any variation the sweep attributes to offered rate. The defensible statement is a range and an order of magnitude, not a ceiling: on this hardware a single controller instance absorbed between 34 and 114 novel flows per second as individual table-miss events, and nothing we measured approached 10^3 .

This bounds the achievable ingest rate, and the bound lies far below the offline throughput figure. The 60,606 flows per second obtained for offline feature extraction and inference was measured with no control channel present. It is not an ingest rate, and it is between two and three orders of magnitude above anything this control plane sustained. Reaching that order of magnitude at the control plane requires that the

TABLE 17. Offered-rate sweep on the OpenFlow testbed, detection without attribution, one 45-second session per row with no background agent. “Achieved” is the synchronous agent’s completed packet-in exchanges per second of wall time, including classification and rule installation. The controller is saturated at every row, including the first, so the table bounds goodput rather than locating the knee. The 2,000, 8,000 and 16,000 rows lie on the same plateau and are in the released results. The 150-second fixed-load session of the text, which also carries benign background traffic, achieved 114 events per second at 3,000 offered and is not comparable row-for-row with these.

Offered (/s)	Achieved (/s)	Ratio	RTT p50 (ms)	RTT p99 (ms)	CPU (%)
500	77	0.155	10.20	38.00	92
1,000	34	0.034	27.46	49.17	89
4,000	34	0.009	27.59	57.12	88
32,000	35	0.001	26.92	48.68	91

flows not arrive as individual packet-in events at all: switch-side aggregation, packet-in rate limiting, and sampled export are the mechanisms, and Section VII-S quantifies what the first of them costs in flow-table occupancy. At the observed 386 bytes per packet-in, 60,000 events per second would also require 23 MB/s on the control channel, which is a separate constraint from controller CPU and one that no aggregation policy removes.

S. FLOW-TABLE OCCUPANCY UNDER MITIGATION

This subsection evaluates which flow-rule aggregation and eviction policies keep a bounded TCAM from overflowing during a flood. The evaluation first tests the premise such an analysis normally rests on, namely that a spoofed flood installs one table entry per spoofed source. On this corpus that premise does not hold, and a different measurement is therefore required.

Across the whole SYN partition, 99.08% of the 3,590,794 flows carry a single source address. In the 40,000-flow window replayed through the testbed there are three distinct source addresses, occupying two /24 networks, and three distinct destination addresses. The flood is not source-diverse at the flow-export level. What is diverse is the port field: the same window contains 29,064 distinct destination ports and 34,624 distinct source ports, which is what makes 29,065 distinct exact-match keys out of 40,000 flows.

Table 18 reports rule installations, peak occupancy, and overflows for six aggregation policies at three table capacities. The consequence of the above is immediate. Exact-match installation saturates the table and overflows continuously, which is the failure mode of concern. Collapsing the source address to a /24 or a /16 changes nothing, to four significant figures, because there is no source diversity to collapse. Wildcarding the source entirely changes nothing either, for the same reason. The aggregation that bounds the table is the one that wildcards the port: a rule matched on source address, destination address, and protocol occupies three entries for the whole window and never overflows at any capacity tested. It installs three rules over the 40,000-flow window against 36,532 for exact matching at the same 4,096-entry capacity, a

TABLE 18. Flow-table occupancy over a 40,000-flow window offered at 3,000 flows/s, under least-recently-used eviction. “Inst./1k” is rule installations per thousand flows; the match is given as the fields retained, everything else being wildcarded. Rows are grouped by behavior: at every capacity the four port-matching policies (exact source, /24, /16, and source wildcarded) agree to every digit printed, as do the two port-wildcarded policies, so one row stands for each group. The complete grid of six aggregations, four eviction policies and three capacities is in the Supplementary Material.

Capacity	Match retained	Inst./1k	Peak	Overflows	Hit rate
1,024	port matched ($\times 4$)	955.6	1,024	37,198	0.044
1,024	port wildcarded ($\times 2$)	0.07	3	0	1.000
4,096	port matched ($\times 4$)	913.3	4,096	32,436	0.087
4,096	port wildcarded ($\times 2$)	0.07	3	0	1.000
16,384	port matched ($\times 4$)	776.6	16,384	14,680	0.223
16,384	port wildcarded ($\times 2$)	0.07	3	0	1.000

reduction of 1.2×10^4 ; the per-thousand-flow column rounds those to 0.07 and 913.3, so the ratio should be read from the counts rather than from the rounded rates. Choice of eviction policy moves the installation count by at most 0.4% under every aggregation, so it is second-order here; the table reports least-recently-used and the full grid is in the released results.

Two qualifications belong with that result. The first is that the reduction is bought with specificity: a rule matched on destination address and protocol drops legitimate traffic to the same host, so port-wildcarded mitigation is an instrument of last resort rather than a default, and the decision of when to escalate to it is a policy question the detector informs but does not settle. The second is that the source-spoofed case, which source-prefix aggregation exists to handle, cannot be observed in this export at all. It is measured separately in the emulated network of Section VII-P, where source addresses are randomized at the packet level and table pressure therefore falls on the source field.

VIII. ABLATION AND COMPARATIVE ANALYSIS

This section compares XAI-SDN against other methods under a single protocol on common hardware. Two analyses often placed in a section of this kind are reported elsewhere in the paper, for reasons of measurement rather than organization. The contribution of the entropy augmentation is quantified in Section VII-F, under the chronological protocol and across five seeds, because a feature ablation measured under a stratified split returns 100.00% accuracy with zero variance on every seed and is therefore uninformative. Leakage robustness is addressed by the audit in Section VII-D, which defines the suspect feature set by measurement rather than by inspection. What remains here is the comparison against other methods.

A. BASELINE COMPARISON

A comparison conducted on a class-balanced subset does not evaluate methods in the setting the proposed detector operates in, and results obtained that way are not informative about deployment. Every method is therefore run twice, on the

same machine, on the CPU, under the same harness: once under protocol A, the full chronological split at the natural class prevalence, and once under protocol B, the subsampled protocol of Section VI-B, which applies the same 100,000-flow training block and 40,000-flow test block to every model at the corpus’s own class ratio. The comparison spans a majority-class baseline, a decision tree, logistic regression, naive Bayes, gradient boosting in both the XGBoost [36] and LightGBM implementations, and support vector machines with linear and RBF kernels [37]. Table 19 reports protocol A, which is the one that describes deployment. One model appears only under protocol B: the RBF support vector machine, whose kernel computation is quadratic in the sample count and which cannot be fitted on 2.5 million flows on a single machine. On the same CPU, harness and test block as every other protocol-B row, but fitted on 40,000 of the 100,000 training flows for the same quadratic reason, it reaches 94.88% macro F1 at a false positive rate of 0.00%, against 97.06% for the detector on the full protocol-B training block. Two models in this comparison are fitted on fewer flows than their peers, and we state both caps rather than presenting either figure as like-for-like; this is the first. We report it here rather than in Table 19 because a figure measured on 40,000 flows does not belong in a column measured on 1,077,238. The second is the linear support vector machine in Table 19, fitted on a capped subsample of 300,000 flows, its liblinear solver being single-threaded and still unconverged after thirty minutes on the full partition; the cap is marked in the table and stated here because a model that cannot be trained at the deployment scale is a deployment fact rather than an inconvenience to be hidden.

Three readings follow, and only the first is favorable to us. The detector and LightGBM reach nearly identical macro F1, 97.88% against 97.89%; we report the two figures rather than calling them equivalent, because no paired significance test was run on this single-run comparison. It attains the lowest false positive rate of any method tested by a factor of four, 0.095% against 0.371% for LightGBM, which over the 1,077,238-flow test partition is 29 false alarms against 113. At an alert volume an analyst has to triage, that difference is the one that decides whether a detector is deployed.

The second reading is that per-flow latency is where the Random Forest is worst. A single-flow decision costs 7.05 ms against 0.26 ms for LightGBM. Our first explanation for that, that unbounded tree depth on 2.5 million flows produces an oversized model, turned out to be wrong when we measured it: Section VII-Q reports that the trees average 46 leaves and the whole forest serializes to 1.46 MB. The gap is a property of the per-call prediction path rather than of the model, which bounds what any change of configuration can recover.

The third is that we make no claim to a uniquely correct choice of classifier. LightGBM reaches the same macro F1 an order of magnitude faster and also admits exact TreeSHAP attribution, so an operator who values throughput over false-alarm volume should prefer it, and we say so in the text rather than leaving it for a reader to notice. What the comparison

supports is correspondingly narrow: at this class ratio, on this corpus, tree ensembles are clearly separated from linear and probabilistic baselines, and the choice within the tree family is a trade between false alarms and latency rather than a matter of accuracy.

B. DEEP-LEARNING BASELINES

To compare against modern deep detectors under a single protocol, five were implemented and run on the CPU under the subsampled protocol that Section VIII-A describes, with the same splits, the same metric definitions and the same timing harness: a 1D convolution over the flow vector, an LSTM and a Transformer encoder over windows of sixteen consecutive flows, a reconstruction autoencoder trained on benign traffic alone, and a two-layer graph convolution over a k -nearest-neighbor graph in flow-feature space. Table 20 reports them.

Four caveats attach to individual rows rather than to the table as a whole, and two of them are departures from protocol B that we state rather than leave in the released results. The sequence models classify a window of sixteen consecutive flows rather than a flow, so their unit of decision is not the unit of enforcement and their test set holds 39,985 windows rather than 40,000 flows. The graph model is transductive over a graph constructed on a 30,000-flow subsample, so single-flow latency is not defined for it without an incremental graph-maintenance scheme we did not implement. The autoencoder is unsupervised, so its operating point is set differently from the rest, and its benign training examples are the 582 benign flows of the whole training partition rather than the 31 that protocol B's training block contains, because 31 examples cannot fit a reconstruction model; that row is therefore trained on more benign data than any other row in either table, which makes its result more favorable than a strict protocol B fit would be and does not change the conclusion drawn from it.

On ranking quality the strongest deep model is close to the tree ensemble and none exceeds it. What separates them is the operating point. At the fixed threshold used throughout this paper the graph model classifies almost nothing as an attack despite a ROC-AUC of 0.9973, because its sigmoid outputs are compressed into a narrow band far below τ . That is a calibration failure rather than a detection failure, and reporting only the fixed-threshold column would misrepresent it, so Section VII-N reports each model at a threshold selected on held-out scores as well.

The autoencoder is the most instructive row. Trained on benign flows only, as the method requires, it assigns *lower* reconstruction error to attack flows than to benign ones: ROC-AUC 0.0073, which inverted is 0.9927. This is not an implementation defect. A randomized SYN flood is far more repetitive than benign traffic, so a model fitted to benign flows reconstructs the attack better than it reconstructs its own training distribution. The premise of reconstruction-based anomaly detection, that the attack is the anomalous class, is false on a corpus where the attack is 97% of the traffic and

structurally uniform. We report it because it bears on any work applying this family of methods to this family of datasets, and because the corpus itself, rather than the method, is what fails here.

C. WHY WE DO NOT TABULATE AGAINST OTHER PAPERS

Placing the figures reported here beside recently published results on CIC-DDoS2019 in a single table would not constitute a controlled comparison, and no such table is presented.

The reasons apply to most cross-paper tables in this literature. Kumar et al. [38] report multi-class identification of the DDoS type, which is a different and harder problem than the binary detection we perform, and an accuracy figure for one is not comparable with an accuracy figure for the other. You et al. [39] evaluate packet-level detection under adversarial perturbation, so both the unit of decision and the threat model differ. Neither uses our partition, our cleaning, or our split protocol, and Section VII-A shows that the last of those alone moves macro F1 by more than two points on identical data, which is larger than the differences such tables are usually read as establishing.

The only ranking claim in this paper is in Section VIII-A, where every method is run by us, on the same data, under the same protocol, on the same hardware.

IX. DISCUSSION

What follows draws out the deployment consequences of the measurements, situates the approach against deep-learning alternatives and against software-defined IoT with its additional constraints, and states what the evidence does not support.

DDoS attacks on SDN controllers threaten infrastructure across enterprise, cloud, and telecommunications networks. A detector fast enough to keep pace with the control channel and transparent enough for an operator to sign off on lowers the practical barrier to deploying machine-learning defenses in these settings, where unexplained automated action is frequently the blocker rather than detection quality.

A. DEEP LEARNING FOR SDN DDOS DETECTION, AND THE TRADE IT OFFERS

Integrating any of the deep detectors of Section VIII-B into a controller is architecturally straightforward, since all of them consume the same exported flow statistics the tree ensemble consumes. The differences that matter are elsewhere. On accuracy the measured gap is small, which is the usual finding on tabular flow features. On latency it is not small and it runs in the tree ensemble's favor. On explainability the asymmetry is structural: TreeSHAP yields exact Shapley values for a tree ensemble in time polynomial in model size [29], whereas a neural detector falls back on a model-agnostic approximation that costs more and is less faithful [8]. None of this makes deep detectors a poor choice in general. It makes them a poor fit for the position this work targets, a detector that runs inside the controller process, decides per flow, and can justify each decision cheaply enough to do so on the alert path.

TABLE 19. Baseline comparison on the full chronological split, 2,513,556 training and 1,077,238 test flows at the natural class prevalence. All figures measured by us on the CPU of Table 5 under one harness. Latency is the median of single-flow inference calls with the estimator forced to serial execution; throughput is measured in batch mode, and the two answer different questions. Bold marks the best value in each column among the three methods that admit exact TreeSHAP attribution, which are the only ones this paper treats as candidates; on accuracy and macro F1 that method is LightGBM rather than ours. [†] trained on a capped subsample of 300,000 flows because the solver does not converge on the full partition.

Method	Acc. (%)	Macro F1 (%)	FPR (%)	Lat. p50 (ms)	Flows/s	Train (s)	Exact XAI
Majority baseline	97.175	49.284	100.000	0.0058	63,672,946	0.1	×
Decision Tree	97.714	66.811	78.421	0.0304	11,775,512	10.9	×
Logistic Regression	98.350	80.873	51.134	0.1086	3,236,092	36.0	×
Naive Bayes	98.590	89.590	0.700	0.1534	2,260,331	2.9	×
XGBoost	99.515	95.802	3.299	0.4168	1,260,010	13.3	✓
Linear SVM [†]	99.641	96.825	3.289	0.1181	3,309,851	6.4	×
LightGBM	99.759	97.888	0.371	0.2551	764,706	5.9	✓
XAI-SDN (RF)	99.758	97.881	0.095	7.0480	27,647	53.3	✓

TABLE 20. Deep-learning baselines under protocol B of Section VI-B, CPU-only, same splits, metric definitions and timing harness as Table 19's protocol-B run. ^s classifies a window of sixteen flows rather than a single flow, over 39,985 windows. ^t transductive over a graph of 20,000 training and 10,000 test nodes drawn from protocol B's blocks, so its figures are computed on 10,000 test flows and single-flow latency is undefined. ^u unsupervised, and the one row whose training data comes from outside protocol B's training block: it is trained on the 582 benign flows of the full training partition, since protocol B's 31 cannot fit a reconstruction model. Its ROC-AUC below 0.5 is discussed in the text. The detector's average precision is printed to five decimal places because it is the only entry that would otherwise round to 1.0000 and be misread as exact. The final row is set in bold because it is this paper's method, not because it is best in any column: four of the five deep baselines have lower single-flow latency, and the parameter counts are not commensurable across model families.

Model	Params	Macro F1 (%)	ROC-AUC	AP	Lat. p50 (ms)
1D-CNN	39,297	81.74	0.8448	0.9908	0.098
LSTM ^s	39,489	91.02	0.9963	0.9999	0.209
Transformer ^s	73,729	81.41	0.9909	0.9997	0.361
Autoencoder ^u	13,544	2.85	0.0073	0.9658	0.041
GNN ^t	7,809	2.80	0.9973	0.9999	n/a
XAI-SDN (RF)	200 trees	97.06	0.9987	0.99996	7.35

B. ADAPTING XAI-SDN TO QOS-AWARE SOFTWARE-DEFINED IOT

Software-defined IoT inherits the SDN control model and adds constraints that change the design problem [20], [40]. Gateways are far weaker than the controller host used here and many are battery powered, so the binding figure is energy per decision rather than peak throughput, which favors a tree ensemble whose inference is comparisons rather than matrix arithmetic. The serialized forest is the constraint instead, and Section VII-Q shows how far it can be shrunk. Window semantics also need re-deriving: a window of N flows spans a very different interval on a sensor network than on a datacenter link, so N should be set from the intended time horizon rather than carried over. The entropy features assume concentration in an attribute is informative, which is weaker on a network where most devices legitimately talk to one endpoint. Finally, mitigation and quality-of-service policy contend for the same flow-table entries: an exact-match rule per attacking flow will evict the entries carrying QoS policy, so the aggregation analysis of Section VII-S is what makes coordinated enforcement feasible. Coupling that to a QoS-aware routing layer, as Gali and Mahamkali propose [40], is the natural extension and is not attempted here.

C. DEPLOYMENT CONSIDERATIONS

Four implications follow from the measurements. Attribution, not classification, is the binding cost: classification sustains 35,715 flows/s and attributing every alert reduces that to 661, so the explanation trigger has to be budgeted rather than left implicit, and Section VII-K gives the frontier to choose a point on. The choice of episode granularity on that frontier is a policy decision with a real cost, since a coarse episode explains the flood rather than the flow. Model size is a material cost: the default configuration serializes to a forest large enough that a single-flow decision costs 7.05 ms, and Section VII-Q shows what bounding tree depth recovers. The rolling entropy engine, by contrast, is a fixed-size buffer plus hash maps and adds no measurable memory pressure. Finally, the threshold can be tuned per deployment to trade false positives against alert volume, but the score it applies to is a forest vote fraction rather than a calibrated posterior, so an operator should treat it as monotone in evidence rather than as a calibrated probability.

D. LIMITATIONS AND THREATS TO VALIDITY

The following bound what this paper establishes. We state them as limits on the claims rather than as directions for future work, because several of them are not resolved by anything we did.

The headline metrics on this benchmark are governed more by the evaluation protocol than by the detector. Section VII-A reports both protocols and quantifies the difference: 97.88% macro F1 under the chronological protocol against 99.95% under the stratified one. Readers comparing against other published results on this corpus should assume that a comparable gap may separate reported figures from what a chronological protocol would yield, unless the paper in question states its protocol.

Dataset-specific artifacts cannot be excluded. The leakage audit in Section VII-D shows that removing every objectively flagged suspect feature leaves performance unchanged, that no feature has disjoint class ranges, and that no test flow has an identical training twin after deduplication. These results are inconsistent with several specific artifact mechanisms. They do not establish that the corpus is free of artifacts, and no feasible experiment on a single synthetic corpus could. This is stated explicitly because an ablation over a small number of features is sometimes taken to confirm the absence of leakage, and it does not.

Attack coverage is partial. The primary evaluation is a SYN flood. Section VII-J extends to other reflection and amplification vectors in the same corpus and to a different capture day, and Section VII-J to two further corpora. Performance degrades under transfer, and we report the degradation rather than the best case. Application-layer attacks, low-rate attacks, and slow-read attacks are not evaluated at all, and nothing here should be read as evidence about them.

The testbed is small. Section VII-R runs the detector inside a controller process exchanging genuine OpenFlow 1.3 messages with switch agents over TCP, and Section VII-P runs it against an emulated network of three hosts and one Open vSwitch datapath under Mininet. Both are real control planes and neither is a network of operational scale. A single switch cannot exhibit multi-datapath rule distribution, controller failover, or the east-west traffic patterns of a datacenter fabric, and the QoS figures we report describe contention at one controller and one 100 Mbit link rather than forwarding-path behavior under production load.

Transferability is bounded by labeled minority traffic, not by labeled volume. Section VII-J1 shows that a hundred labeled flows from one target network are enough to reach 0.9964 macro F1, while fifty thousand from another are not enough to reach 0.90. The difference is the number of benign examples the target's own capture contains. An operator cannot infer the labeling cost of deployment from the flow count alone, and nothing measured here predicts it.

The adaptive adversary is out of scope. The threat model assumes an attacker who floods but does not optimize against the detector. An adversary who knows the feature set can rotate spoofed sources to hold per-window entropy near its benign value, and the distributional evidence degrades accordingly. We have not measured what that evasion costs the attacker in attack effectiveness, which is the quantity that would determine whether the defense remains useful under adaptation. This is the most consequential gap in the paper.

Budgeted alerts do not carry an exact attribution of their own score. Under any policy that admits fewer alerts than it receives, the alerts the budget declines carry the attribution of their episode representative, $\phi_{\rho(i)}$ in Eq. (4). That vector satisfies the efficiency property of Eq. (3) for the representative flow and not, in general, for the flow it is attached to, so it identifies why the episode was blocked rather than why this flow was scored as it was. At the per-service granularity that recovers the classification rate, that applies to all but two of 59,998 alerts. An operator who needs a per-flow answer must tighten the granularity or re-explain the flow on demand, and the frontier in Section VII-K is the cost of doing so. We flag this because the unbudgeted reading, in which every alert carries its own exact attribution, is the one a reader would otherwise carry over from the abstract.

Detector confidence is uncalibrated. The score is a forest vote fraction. It is monotone in evidence and adequate for thresholding, but it is not a calibrated posterior, and the alert payload presents it to an operator as though it were interpretable on its own scale. Bayesian treatment of this quantity [31] would give the operator something better to act on at borderline scores, and is not attempted here.

X. CONCLUSION

XAI-SDN is a controller-embedded DDoS detector that augments flow statistics with windowed Shannon entropy maintained at expected amortized $\mathcal{O}(1)$ cost, classifies with a Random Forest, and attributes alerts with exact TreeSHAP under an explanation budget, every admitted alert receiving an attribution of its own score and every other resolving to its episode representative. Under a chronological split of the CIC-DDoS2019 SYN partition it reaches 99.76% accuracy and 97.88% macro F1 over 1,077,238 held-out flows, at a false positive rate of 0.095%, the lowest of any method we evaluated.

Three further results bear more directly on deployment than the detection figure does. Explanation is affordable only if its trigger is budgeted, since attributing every positive prediction collapses sustained throughput when almost all traffic is hostile, and the throughput-versus-coverage frontier we report lets an operator choose a point on it rather than discover the cost in production. Control-plane cost is not inference cost: measured inside a live OpenFlow controller, the binding constraints are the rate at which table-miss events can be ingested and the rate at which mitigation rules exhaust a bounded flow table, neither of which appears in an offline throughput number. And the evaluation protocol moves the headline metrics on this benchmark further than any architectural choice we examined, which is a caution that applies well beyond this paper.

The paper is equally explicit about what the evidence does not support. The entropy augmentation does not improve in-distribution detection on this corpus, and buys robustness under shift only where its full feature set is computable. The detector does not transfer to another corpus without retraining, and on one of the two target corpora evaluated the

source model is actively harmful, so that training on a hundred labeled target flows yields better performance than transferring it. No experiment available to us can exclude dataset-specific artifacts. Section IX-D states these as limits on the claims rather than as future work. The work we consider most pressing follows from them: quantifying what evasion costs an adaptive adversary who shapes per-window distributions, calibrating the detector score so that borderline alerts carry usable confidence, and evaluating on captures whose benign traffic is not confined to a single interval of the recording.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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DATA AVAILABILITY

The datasets, implementation code, trained-model artifacts, per-seed manifests, and experimental configuration files supporting this study are publicly available at: <https://github.com/adeliusa486/XAI-SDN>

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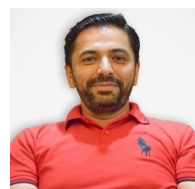
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